

diabetes

December 6, 2025

```
[1]: # Import necessary libraries
import optuna
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Load the Pima Indian Diabetes dataset from sklearn
# Note: Scikit-learn's built-in 'load_diabetes' is a regression dataset.
# We will load the actual diabetes dataset from an external source
import pandas as pd

# Load the Pima Indian Diabetes dataset (from UCI repository)
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/
↳pima-indians-diabetes.data.csv"
columns = ['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', '
↳'Insulin', 'BMI',
           'DiabetesPedigreeFunction', 'Age', 'Outcome']

# Load the dataset
df = pd.read_csv(url, names=columns)

df.head()
```

```
[1]:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
[2]: import numpy as np

# Replace zero values with NaN in columns where zero is not a valid value
cols_with_missing_vals = ['Glucose', 'BloodPressure', 'SkinThickness',
    ↳ 'Insulin', 'BMI']
df[cols_with_missing_vals] = df[cols_with_missing_vals].replace(0, np.nan)

# Impute the missing values with the mean of the respective column
df.fillna(df.mean(), inplace=True)

# Check if there are any remaining missing values
print(df.isnull().sum())
```

```
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction  0
Age              0
Outcome          0
dtype: int64
```

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[3]: # Split into features (X) and target (y)
X = df.drop('Outcome', axis=1)
y = df['Outcome']

# Split data into training and test sets (70% train, 30% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
    ↳ random_state=42)

# Optional: Scale the data for better model performance
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Check the shape of the data
print(f'Training set shape: {X_train.shape}')
print(f'Test set shape: {X_test.shape}')
```

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Training set shape: (537, 8)
Test set shape: (231, 8)
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[4]: from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import cross_val_score
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# Define the objective function
def objective(trial):
    # Suggest values for the hyperparameters
    n_estimators = trial.suggest_int('n_estimators', 50, 200)
    max_depth = trial.suggest_int('max_depth', 3, 20)

    # Create the RandomForestClassifier with suggested hyperparameters
    model = RandomForestClassifier(
        n_estimators=n_estimators,
        max_depth=max_depth,
        random_state=42
    )

    # Perform 3-fold cross-validation and calculate accuracy
    score = cross_val_score(model, X_train, y_train, cv=3, scoring='accuracy').
    ↪mean()

    return score # Return the accuracy score for Optuna to maximize

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[5]: # Create a study object and optimize the objective function
study = optuna.create_study(direction='maximize', sampler=optuna.samplers.
    ↪TPESampler()) # We aim to maximize accuracy
study.optimize(objective, n_trials=50) # Run 50 trials to find the best
    ↪hyperparameters

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[I 2025-12-06 14:51:24,648] A new study created in memory with name: no-
name-8bec122c-66e4-4da5-82a4-bf4a2d86a1dc
[I 2025-12-06 14:51:25,554] Trial 0 finished with value: 0.756052141527002 and
parameters: {'n_estimators': 120, 'max_depth': 3}. Best is trial 0 with value:
0.756052141527002.
[I 2025-12-06 14:51:26,393] Trial 1 finished with value: 0.7728119180633147 and
parameters: {'n_estimators': 116, 'max_depth': 14}. Best is trial 1 with value:
0.7728119180633147.
[I 2025-12-06 14:51:27,121] Trial 2 finished with value: 0.7635009310986964 and
parameters: {'n_estimators': 104, 'max_depth': 4}. Best is trial 1 with value:
0.7728119180633147.
[I 2025-12-06 14:51:27,741] Trial 3 finished with value: 0.7765363128491621 and
parameters: {'n_estimators': 77, 'max_depth': 7}. Best is trial 3 with value:
0.7765363128491621.
[I 2025-12-06 14:51:28,886] Trial 4 finished with value: 0.7765363128491621 and
parameters: {'n_estimators': 120, 'max_depth': 18}. Best is trial 3 with value:
0.7765363128491621.
[I 2025-12-06 14:51:29,396] Trial 5 finished with value: 0.7635009310986964 and
parameters: {'n_estimators': 75, 'max_depth': 4}. Best is trial 3 with value:
0.7765363128491621.
[I 2025-12-06 14:51:30,156] Trial 6 finished with value: 0.7616387337057727 and
parameters: {'n_estimators': 112, 'max_depth': 9}. Best is trial 3 with value:

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0.7765363128491621.

[I 2025-12-06 14:51:31,351] Trial 7 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 179, 'max_depth': 18}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:32,652] Trial 8 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 184, 'max_depth': 19}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:33,824] Trial 9 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 140, 'max_depth': 20}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:34,625] Trial 10 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 56, 'max_depth': 9}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:36,585] Trial 11 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 84, 'max_depth': 14}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:39,018] Trial 12 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 151, 'max_depth': 8}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:39,424] Trial 13 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 51, 'max_depth': 15}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:40,551] Trial 14 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 88, 'max_depth': 7}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:41,723] Trial 15 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 157, 'max_depth': 11}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:42,397] Trial 16 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 96, 'max_depth': 17}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:42,934] Trial 17 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 70, 'max_depth': 6}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:44,169] Trial 18 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 135, 'max_depth': 11}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:46,332] Trial 19 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 160, 'max_depth': 16}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:46,812] Trial 20 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 65, 'max_depth': 13}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:48,256] Trial 21 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 173, 'max_depth': 20}. Best is trial 3 with value: 0.7765363128491621.

[I 2025-12-06 14:51:50,035] Trial 22 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 194, 'max_depth': 18}. Best is trial 3 with value:

0.7765363128491621.

[I 2025-12-06 14:51:53,443] Trial 23 finished with value: 0.7821229050279329 and parameters: {'n_estimators': 197, 'max_depth': 19}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:51:54,984] Trial 24 finished with value: 0.7765363128491619 and parameters: {'n_estimators': 127, 'max_depth': 12}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:51:57,117] Trial 25 finished with value: 0.7802607076350093 and parameters: {'n_estimators': 198, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:51:59,110] Trial 26 finished with value: 0.7783985102420857 and parameters: {'n_estimators': 194, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:01,200] Trial 27 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 200, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:03,288] Trial 28 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 192, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:04,721] Trial 29 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 170, 'max_depth': 17}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:06,561] Trial 30 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 186, 'max_depth': 19}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:09,252] Trial 31 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 198, 'max_depth': 15}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:12,273] Trial 32 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 199, 'max_depth': 15}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:14,030] Trial 33 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 177, 'max_depth': 13}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:15,755] Trial 34 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 188, 'max_depth': 17}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:17,500] Trial 35 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 166, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:20,002] Trial 36 finished with value: 0.7765363128491621 and parameters: {'n_estimators': 199, 'max_depth': 19}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:21,398] Trial 37 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 185, 'max_depth': 14}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:23,394] Trial 38 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 180, 'max_depth': 18}. Best is trial 23 with value:

0.7821229050279329.

[I 2025-12-06 14:52:25,461] Trial 39 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 147, 'max_depth': 13}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:26,732] Trial 40 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 165, 'max_depth': 20}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:28,054] Trial 41 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 192, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:29,365] Trial 42 finished with value: 0.7783985102420857 and parameters: {'n_estimators': 194, 'max_depth': 16}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:30,727] Trial 43 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 200, 'max_depth': 17}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:31,997] Trial 44 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 178, 'max_depth': 15}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:33,519] Trial 45 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 191, 'max_depth': 18}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:35,184] Trial 46 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 183, 'max_depth': 19}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:36,338] Trial 47 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 175, 'max_depth': 14}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:37,778] Trial 48 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 193, 'max_depth': 12}. Best is trial 23 with value: 0.7821229050279329.

[I 2025-12-06 14:52:39,800] Trial 49 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 184, 'max_depth': 10}. Best is trial 23 with value: 0.7821229050279329.

```
[6]: # Print the best result
      print(f'Best trial accuracy: {study.best_trial.value}')
      print(f'Best hyperparameters: {study.best_trial.params}')
```

Best trial accuracy: 0.7821229050279329

Best hyperparameters: {'n_estimators': 197, 'max_depth': 19}

```
[7]: from sklearn.metrics import accuracy_score

      # Train a RandomForestClassifier using the best hyperparameters from Optuna
      best_model = RandomForestClassifier(**study.best_trial.params, random_state=42)
```

```

# Fit the model to the training data
best_model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = best_model.predict(X_test)

# Calculate the accuracy on the test set
test_accuracy = accuracy_score(y_test, y_pred)

# Print the test accuracy
print(f'Test Accuracy with best hyperparameters: {test_accuracy:.2f}')

```

Test Accuracy with best hyperparameters: 0.74

```

[8]: from sklearn.ensemble import RandomForestClassifier
      from sklearn.model_selection import cross_val_score

      # Define the objective function
      def objective(trial):
          # Suggest values for the hyperparameters
          n_estimators = trial.suggest_int('n_estimators', 50, 200)
          max_depth = trial.suggest_int('max_depth', 3, 20)

          # Create the RandomForestClassifier with suggested hyperparameters
          model = RandomForestClassifier(
              n_estimators=n_estimators,
              max_depth=max_depth,
              random_state=42
          )

          # Perform 3-fold cross-validation and calculate accuracy
          score = cross_val_score(model, X_train, y_train, cv=3, scoring='accuracy').
          ↪mean()

          return score # Return the accuracy score for Optuna to maximize

```

```

[9]: study = optuna.create_study(direction='maximize', sampler=optuna.samplers.
      ↪RandomSampler()) # We aim to maximize accuracy
      study.optimize(objective, n_trials=50) # Run 50 trials to find the best
      ↪hyperparameters

```

[I 2025-12-06 14:52:40,837] A new study created in memory with name: no-name-e9172498-ca0c-4b18-9d89-f6f3a51de054

[I 2025-12-06 14:52:42,506] Trial 0 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 170, 'max_depth': 4}. Best is trial 0 with value: 0.7635009310986964.

[I 2025-12-06 14:52:44,172] Trial 1 finished with value: 0.756052141527002 and

parameters: {'n_estimators': 167, 'max_depth': 3}. Best is trial 0 with value: 0.7635009310986964.

[I 2025-12-06 14:52:45,301] Trial 2 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 101, 'max_depth': 11}. Best is trial 2 with value: 0.7653631284916201.

[I 2025-12-06 14:52:45,774] Trial 3 finished with value: 0.7541899441340782 and parameters: {'n_estimators': 55, 'max_depth': 10}. Best is trial 2 with value: 0.7653631284916201.

[I 2025-12-06 14:52:46,357] Trial 4 finished with value: 0.7523277467411545 and parameters: {'n_estimators': 86, 'max_depth': 3}. Best is trial 2 with value: 0.7653631284916201.

[I 2025-12-06 14:52:47,624] Trial 5 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 175, 'max_depth': 17}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:48,876] Trial 6 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 166, 'max_depth': 20}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:49,489] Trial 7 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 84, 'max_depth': 20}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:50,996] Trial 8 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 172, 'max_depth': 17}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:52,140] Trial 9 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 128, 'max_depth': 6}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:52,610] Trial 10 finished with value: 0.7523277467411545 and parameters: {'n_estimators': 62, 'max_depth': 3}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:53,382] Trial 11 finished with value: 0.7560521415270017 and parameters: {'n_estimators': 100, 'max_depth': 3}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:54,808] Trial 12 finished with value: 0.756052141527002 and parameters: {'n_estimators': 121, 'max_depth': 3}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:56,520] Trial 13 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 127, 'max_depth': 4}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:57,878] Trial 14 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 108, 'max_depth': 15}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:58,689] Trial 15 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 73, 'max_depth': 15}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:52:59,451] Trial 16 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 82, 'max_depth': 17}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:00,681] Trial 17 finished with value: 0.756052141527002 and

parameters: {'n_estimators': 173, 'max_depth': 4}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:01,423] Trial 18 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 88, 'max_depth': 20}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:02,270] Trial 19 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 123, 'max_depth': 3}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:03,072] Trial 20 finished with value: 0.756052141527002 and parameters: {'n_estimators': 119, 'max_depth': 3}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:04,175] Trial 21 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 159, 'max_depth': 6}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:04,871] Trial 22 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 96, 'max_depth': 13}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:05,893] Trial 23 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 152, 'max_depth': 19}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:07,033] Trial 24 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 178, 'max_depth': 6}. Best is trial 5 with value: 0.7746741154562384.

[I 2025-12-06 14:53:07,793] Trial 25 finished with value: 0.7783985102420856 and parameters: {'n_estimators': 111, 'max_depth': 16}. Best is trial 25 with value: 0.7783985102420856.

[I 2025-12-06 14:53:09,117] Trial 26 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 192, 'max_depth': 8}. Best is trial 25 with value: 0.7783985102420856.

[I 2025-12-06 14:53:10,027] Trial 27 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 80, 'max_depth': 14}. Best is trial 25 with value: 0.7783985102420856.

[I 2025-12-06 14:53:11,665] Trial 28 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 148, 'max_depth': 5}. Best is trial 25 with value: 0.7783985102420856.

[I 2025-12-06 14:53:13,257] Trial 29 finished with value: 0.7821229050279329 and parameters: {'n_estimators': 119, 'max_depth': 19}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:14,199] Trial 30 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 109, 'max_depth': 4}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:15,223] Trial 31 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 118, 'max_depth': 6}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:16,269] Trial 32 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 138, 'max_depth': 20}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:17,658] Trial 33 finished with value: 0.7616387337057727 and

parameters: {'n_estimators': 193, 'max_depth': 9}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:18,196] Trial 34 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 57, 'max_depth': 11}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:18,704] Trial 35 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 58, 'max_depth': 11}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:20,143] Trial 36 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 169, 'max_depth': 5}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:21,183] Trial 37 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 100, 'max_depth': 16}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:22,858] Trial 38 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 173, 'max_depth': 20}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:24,492] Trial 39 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 174, 'max_depth': 14}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:25,267] Trial 40 finished with value: 0.756052141527002 and parameters: {'n_estimators': 119, 'max_depth': 3}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:26,832] Trial 41 finished with value: 0.7541899441340782 and parameters: {'n_estimators': 164, 'max_depth': 3}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:28,036] Trial 42 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 110, 'max_depth': 14}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:29,383] Trial 43 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 157, 'max_depth': 6}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:30,315] Trial 44 finished with value: 0.7746741154562383 and parameters: {'n_estimators': 135, 'max_depth': 15}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:30,965] Trial 45 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 96, 'max_depth': 7}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:31,551] Trial 46 finished with value: 0.7597765363128491 and parameters: {'n_estimators': 84, 'max_depth': 20}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:32,296] Trial 47 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 109, 'max_depth': 8}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:33,422] Trial 48 finished with value: 0.7635009310986964 and parameters: {'n_estimators': 171, 'max_depth': 8}. Best is trial 29 with value: 0.7821229050279329.

[I 2025-12-06 14:53:34,015] Trial 49 finished with value: 0.7653631284916201 and

parameters: {'n_estimators': 82, 'max_depth': 17}. Best is trial 29 with value: 0.7821229050279329.

```
[10]: # Print the best result
print(f'Best trial accuracy: {study.best_trial.value}')
print(f'Best hyperparameters: {study.best_trial.params}')
```

Best trial accuracy: 0.7821229050279329
Best hyperparameters: {'n_estimators': 119, 'max_depth': 19}

```
[11]: from sklearn.metrics import accuracy_score

# Train a RandomForestClassifier using the best hyperparameters from Optuna
best_model = RandomForestClassifier(**study.best_trial.params, random_state=42)

# Fit the model to the training data
best_model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = best_model.predict(X_test)

# Calculate the accuracy on the test set
test_accuracy = accuracy_score(y_test, y_pred)

# Print the test accuracy
print(f'Test Accuracy with best hyperparameters: {test_accuracy:.2f}')
```

Test Accuracy with best hyperparameters: 0.74

```
[12]: search_space = {
    'n_estimators': [50, 100, 150, 200],
    'max_depth': [5, 10, 15, 20]
}
```

```
[13]: # Create a study and optimize it using GridSampler
study = optuna.create_study(direction='maximize', sampler=optuna.samplers.
    ↪GridSampler(search_space))
study.optimize(objective)
```

[I 2025-12-06 14:53:34,567] A new study created in memory with name: no-name-07b23ebb-4bb6-4065-ac78-99bd70cd2b44

[I 2025-12-06 14:53:35,400] Trial 0 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 100, 'max_depth': 5}. Best is trial 0 with value: 0.7690875232774674.

[I 2025-12-06 14:53:37,387] Trial 1 finished with value: 0.7672253258845437 and parameters: {'n_estimators': 150, 'max_depth': 10}. Best is trial 0 with value: 0.7690875232774674.

[I 2025-12-06 14:53:37,750] Trial 2 finished with value: 0.7728119180633147 and

parameters: {'n_estimators': 50, 'max_depth': 15}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:38,446] Trial 3 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 100, 'max_depth': 15}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:39,111] Trial 4 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 100, 'max_depth': 20}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:39,460] Trial 5 finished with value: 0.7579143389199254 and parameters: {'n_estimators': 50, 'max_depth': 10}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:40,442] Trial 6 finished with value: 0.7653631284916201 and parameters: {'n_estimators': 150, 'max_depth': 5}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:41,629] Trial 7 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 150, 'max_depth': 20}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:43,289] Trial 8 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 150, 'max_depth': 15}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:45,184] Trial 9 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 200, 'max_depth': 10}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:46,677] Trial 10 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 200, 'max_depth': 20}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:48,221] Trial 11 finished with value: 0.7728119180633147 and parameters: {'n_estimators': 200, 'max_depth': 15}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:49,875] Trial 12 finished with value: 0.7690875232774674 and parameters: {'n_estimators': 200, 'max_depth': 5}. Best is trial 2 with value: 0.7728119180633147.

[I 2025-12-06 14:53:50,256] Trial 13 finished with value: 0.7746741154562384 and parameters: {'n_estimators': 50, 'max_depth': 5}. Best is trial 13 with value: 0.7746741154562384.

[I 2025-12-06 14:53:51,230] Trial 14 finished with value: 0.7616387337057727 and parameters: {'n_estimators': 100, 'max_depth': 10}. Best is trial 13 with value: 0.7746741154562384.

[I 2025-12-06 14:53:52,196] Trial 15 finished with value: 0.7709497206703911 and parameters: {'n_estimators': 50, 'max_depth': 20}. Best is trial 13 with value: 0.7746741154562384.

```
[14]: # Print the best result
print(f'Best trial accuracy: {study.best_trial.value}')
print(f'Best hyperparameters: {study.best_trial.params}')
```

Best trial accuracy: 0.7746741154562384
Best hyperparameters: {'n_estimators': 50, 'max_depth': 5}

```
[15]: from sklearn.metrics import accuracy_score

# Train a RandomForestClassifier using the best hyperparameters from Optuna
best_model = RandomForestClassifier(**study.best_trial.params, random_state=42)

# Fit the model to the training data
best_model.fit(X_train, y_train)

# Make predictions on the test set
y_pred = best_model.predict(X_test)

# Calculate the accuracy on the test set
test_accuracy = accuracy_score(y_test, y_pred)

# Print the test accuracy
print(f'Test Accuracy with best hyperparameters: {test_accuracy:.2f}')
```

Test Accuracy with best hyperparameters: 0.74

```
[16]: # For visualizations
from optuna.visualization import plot_optimization_history,
    plot_parallel_coordinate, plot_slice, plot_contour, plot_param_importances
```

```
[17]: # 1. Optimization History
plot_optimization_history(study).show()
```

```
[18]: # 2. Parallel Coordinates Plot
plot_parallel_coordinate(study).show()
```

```
[19]: # 3. Slice Plot
plot_slice(study).show()
```

```
[20]: # 4. Contour Plot
plot_contour(study).show()
```

```
[21]: # 5. Hyperparameter Importance
plot_param_importances(study).show()
```

```
[22]: # Importing the required libraries
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.svm import SVC
```

```
[23]: # Define the objective function for Optuna
def objective(trial):
    # Choose the algorithm to tune
    classifier_name = trial.suggest_categorical('classifier', ['SVM',
        'RandomForest', 'GradientBoosting'])
```

```

if classifier_name == 'SVM':
    # SVM hyperparameters
    c = trial.suggest_float('C', 0.1, 100, log=True)
    kernel = trial.suggest_categorical('kernel', ['linear', 'rbf', 'poly', '
↪sigmoid'])
    gamma = trial.suggest_categorical('gamma', ['scale', 'auto'])

    model = SVC(C=c, kernel=kernel, gamma=gamma, random_state=42)

elif classifier_name == 'RandomForest':
    # Random Forest hyperparameters
    n_estimators = trial.suggest_int('n_estimators', 50, 300)
    max_depth = trial.suggest_int('max_depth', 3, 20)
    min_samples_split = trial.suggest_int('min_samples_split', 2, 10)
    min_samples_leaf = trial.suggest_int('min_samples_leaf', 1, 10)
    bootstrap = trial.suggest_categorical('bootstrap', [True, False])

    model = RandomForestClassifier(
        n_estimators=n_estimators,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        min_samples_leaf=min_samples_leaf,
        bootstrap=bootstrap,
        random_state=42
    )

elif classifier_name == 'GradientBoosting':
    # Gradient Boosting hyperparameters
    n_estimators = trial.suggest_int('n_estimators', 50, 300)
    learning_rate = trial.suggest_float('learning_rate', 0.01, 0.3,
↪log=True)
    max_depth = trial.suggest_int('max_depth', 3, 20)
    min_samples_split = trial.suggest_int('min_samples_split', 2, 10)
    min_samples_leaf = trial.suggest_int('min_samples_leaf', 1, 10)

    model = GradientBoostingClassifier(
        n_estimators=n_estimators,
        learning_rate=learning_rate,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        min_samples_leaf=min_samples_leaf,
        random_state=42
    )

# Perform cross-validation and return the mean accuracy

```

```

    score = cross_val_score(model, X_train, y_train, cv=3, scoring='accuracy').
    ↪mean()
    return score

```

```

[24]: # Create a study and optimize it using CmaEsSampler
study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=100)

```

```

[I 2025-12-06 14:53:59,944] A new study created in memory with name: no-name-
af85b679-7b23-4b3f-95c3-66ed88eb0e1f
[I 2025-12-06 14:54:03,194] Trial 0 finished with value: 0.7690875232774674 and
parameters: {'classifier': 'RandomForest', 'n_estimators': 174, 'max_depth': 10,
'min_samples_split': 4, 'min_samples_leaf': 3, 'bootstrap': True}. Best is trial
0 with value: 0.7690875232774674.
[I 2025-12-06 14:54:03,257] Trial 1 finished with value: 0.7709497206703911 and
parameters: {'classifier': 'SVM', 'C': 1.5671083595814452, 'kernel': 'rbf',
'gamma': 'auto'}. Best is trial 1 with value: 0.7709497206703911.
[I 2025-12-06 14:54:06,632] Trial 2 finished with value: 0.7560521415270017 and
parameters: {'classifier': 'RandomForest', 'n_estimators': 270, 'max_depth': 9,
'min_samples_split': 5, 'min_samples_leaf': 10, 'bootstrap': True}. Best is
trial 1 with value: 0.7709497206703911.
[I 2025-12-06 14:54:06,729] Trial 3 finished with value: 0.7188081936685289 and
parameters: {'classifier': 'SVM', 'C': 44.811487918360406, 'kernel': 'rbf',
'gamma': 'auto'}. Best is trial 1 with value: 0.7709497206703911.
[I 2025-12-06 14:54:06,920] Trial 4 finished with value: 0.7858472998137801 and
parameters: {'classifier': 'SVM', 'C': 23.772036773578954, 'kernel': 'linear',
'gamma': 'auto'}. Best is trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:06,976] Trial 5 finished with value: 0.7150837988826816 and
parameters: {'classifier': 'SVM', 'C': 0.17965071844081174, 'kernel': 'poly',
'gamma': 'auto'}. Best is trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:10,212] Trial 6 finished with value: 0.7318435754189944 and
parameters: {'classifier': 'GradientBoosting', 'n_estimators': 191,
'learning_rate': 0.1834200708691115, 'max_depth': 12, 'min_samples_split': 7,
'min_samples_leaf': 9}. Best is trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:12,320] Trial 7 finished with value: 0.7374301675977654 and
parameters: {'classifier': 'GradientBoosting', 'n_estimators': 110,
'learning_rate': 0.2823919675478522, 'max_depth': 13, 'min_samples_split': 2,
'min_samples_leaf': 9}. Best is trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:13,280] Trial 8 finished with value: 0.7783985102420856 and
parameters: {'classifier': 'RandomForest', 'n_estimators': 115, 'max_depth': 8,
'min_samples_split': 9, 'min_samples_leaf': 6, 'bootstrap': False}. Best is
trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:17,574] Trial 9 finished with value: 0.7299813780260708 and
parameters: {'classifier': 'GradientBoosting', 'n_estimators': 139,
'learning_rate': 0.2645831274264107, 'max_depth': 17, 'min_samples_split': 7,
'min_samples_leaf': 6}. Best is trial 4 with value: 0.7858472998137801.
[I 2025-12-06 14:54:17,820] Trial 10 finished with value: 0.7858472998137801 and

```

parameters: {'classifier': 'SVM', 'C': 51.23848449173574, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:18,213] Trial 11 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 72.50386756520165, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:18,330] Trial 12 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 12.037845233891504, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:19,932] Trial 13 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 9.818720239233986, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:20,263] Trial 14 finished with value: 0.6945996275605214 and parameters: {'classifier': 'SVM', 'C': 94.48462158406252, 'kernel': 'sigmoid', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:21,077] Trial 15 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 14.619869535073432, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:21,344] Trial 16 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 2.208714223216749, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:21,387] Trial 17 finished with value: 0.690875232774674 and parameters: {'classifier': 'SVM', 'C': 24.966021096457002, 'kernel': 'sigmoid', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:21,861] Trial 18 finished with value: 0.7392923649906891 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 53, 'learning_rate': 0.011301198966018467, 'max_depth': 4, 'min_samples_split': 10, 'min_samples_leaf': 1}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:23,700] Trial 19 finished with value: 0.7616387337057727 and parameters: {'classifier': 'RandomForest', 'n_estimators': 291, 'max_depth': 20, 'min_samples_split': 2, 'min_samples_leaf': 4, 'bootstrap': False}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:23,780] Trial 20 finished with value: 0.7225325884543762 and parameters: {'classifier': 'SVM', 'C': 5.714043538797612, 'kernel': 'poly', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:24,183] Trial 21 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 91.54349070347081, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:24,378] Trial 22 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 35.17234653948882, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:24,605] Trial 23 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 45.25786359683838, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:25,064] Trial 24 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 98.45158009472121, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 4 with value: 0.7858472998137801.

[I 2025-12-06 14:54:25,120] Trial 25 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.7172043462125243, 'kernel': 'linear',

'gamma': 'scale'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:25,164] Trial 26 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.6022537409348706, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:25,380] Trial 27 finished with value: 0.750465549348231 and parameters: {'classifier': 'SVM', 'C': 0.9004904851504701, 'kernel': 'sigmoid', 'gamma': 'scale'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:27,036] Trial 28 finished with value: 0.7635009310986964 and parameters: {'classifier': 'RandomForest', 'n_estimators': 228, 'max_depth': 3, 'min_samples_split': 8, 'min_samples_leaf': 1, 'bootstrap': True}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:28,168] Trial 29 finished with value: 0.7690875232774674 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 71, 'learning_rate': 0.021811528489707802, 'max_depth': 16, 'min_samples_split': 4, 'min_samples_leaf': 7}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:29,693] Trial 30 finished with value: 0.7839851024208566 and parameters: {'classifier': 'RandomForest', 'n_estimators': 250, 'max_depth': 6, 'min_samples_split': 10, 'min_samples_leaf': 3, 'bootstrap': False}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:29,776] Trial 31 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 4.836196689899744, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:29,844] Trial 32 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.274115178095562, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:29,909] Trial 33 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.2152513238676899, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 25 with value: 0.7858472998137803.

[I 2025-12-06 14:54:29,963] Trial 34 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.1583538575619416, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:30,035] Trial 35 finished with value: 0.7430167597765363 and parameters: {'classifier': 'SVM', 'C': 0.1211397050879638, 'kernel': 'rbf', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:30,093] Trial 36 finished with value: 0.7262569832402234 and parameters: {'classifier': 'SVM', 'C': 0.30040382639856766, 'kernel': 'poly', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:30,158] Trial 37 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.35622959501998536, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:30,222] Trial 38 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.23446651651062447, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:32,346] Trial 39 finished with value: 0.7597765363128491 and parameters: {'classifier': 'RandomForest', 'n_estimators': 208, 'max_depth': 15, 'min_samples_split': 6, 'min_samples_leaf': 8, 'bootstrap': True}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:37,341] Trial 40 finished with value: 0.7392923649906891 and

parameters: {'classifier': 'GradientBoosting', 'n_estimators': 157, 'learning_rate': 0.06454578386706465, 'max_depth': 20, 'min_samples_split': 3, 'min_samples_leaf': 4}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:37,398] Trial 41 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.27128442403508685, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 34 with value: 0.7877094972067038.

[I 2025-12-06 14:54:37,427] Trial 42 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.11063344610550901, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:37,482] Trial 43 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.10320949042668802, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:37,542] Trial 44 finished with value: 0.7132216014897579 and parameters: {'classifier': 'SVM', 'C': 0.10607434311210184, 'kernel': 'rbf', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:37,585] Trial 45 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.49952103544295834, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:37,642] Trial 46 finished with value: 0.7523277467411545 and parameters: {'classifier': 'SVM', 'C': 0.15226148127749775, 'kernel': 'rbf', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:37,697] Trial 47 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 1.0450554135585621, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,307] Trial 48 finished with value: 0.7486033519553073 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 300, 'learning_rate': 0.0747800244429118, 'max_depth': 6, 'min_samples_split': 6, 'min_samples_leaf': 2}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,360] Trial 49 finished with value: 0.7132216014897579 and parameters: {'classifier': 'SVM', 'C': 0.15061321747311598, 'kernel': 'poly', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,409] Trial 50 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.100747707851439, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,443] Trial 51 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.11103754433143441, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,494] Trial 52 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.10920745279971018, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,535] Trial 53 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.1026583529982685, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,580] Trial 54 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.1642302176947284, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:41,620] Trial 55 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.10310963933993089, 'kernel': 'linear',

'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:42,162] Trial 56 finished with value: 0.7728119180633147 and parameters: {'classifier': 'RandomForest', 'n_estimators': 85, 'max_depth': 18, 'min_samples_split': 8, 'min_samples_leaf': 5, 'bootstrap': False}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:42,213] Trial 57 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.40839989931316156, 'kernel': 'linear', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:42,267] Trial 58 finished with value: 0.7709497206703911 and parameters: {'classifier': 'SVM', 'C': 0.17394152948423225, 'kernel': 'sigmoid', 'gamma': 'scale'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:42,323] Trial 59 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.13879797917321132, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,384] Trial 60 finished with value: 0.7579143389199255 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 133, 'learning_rate': 0.029064006668097744, 'max_depth': 15, 'min_samples_split': 4, 'min_samples_leaf': 10}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,443] Trial 61 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.1443997632791171, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,497] Trial 62 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.18041616518926581, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,565] Trial 63 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.1413657461005914, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,631] Trial 64 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.14835446234553032, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,890] Trial 65 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.14608732024281088, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,948] Trial 66 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.2084857710184607, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:44,995] Trial 67 finished with value: 0.7113594040968342 and parameters: {'classifier': 'SVM', 'C': 0.1389976110431635, 'kernel': 'poly', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:45,025] Trial 68 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.4067391093449276, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:47,827] Trial 69 finished with value: 0.750465549348231 and parameters: {'classifier': 'RandomForest', 'n_estimators': 224, 'max_depth': 13, 'min_samples_split': 5, 'min_samples_leaf': 7, 'bootstrap': True}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:47,894] Trial 70 finished with value: 0.7616387337057727 and parameters: {'classifier': 'SVM', 'C': 0.24435235479340645, 'kernel': 'sigmoid',

'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:47,942] Trial 71 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.1474577113621228, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,011] Trial 72 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.138124794881839, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,086] Trial 73 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.21122867702970277, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,172] Trial 74 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.3053205505956338, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,259] Trial 75 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.19590602778456095, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,350] Trial 76 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.14228144730417516, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:48,444] Trial 77 finished with value: 0.7523277467411545 and parameters: {'classifier': 'SVM', 'C': 0.13430535264191515, 'kernel': 'rbf', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,748] Trial 78 finished with value: 0.750465549348231 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 252, 'learning_rate': 0.13626762061905975, 'max_depth': 6, 'min_samples_split': 9, 'min_samples_leaf': 5}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,794] Trial 79 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 1.7509816001973564, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,833] Trial 80 finished with value: 0.7839851024208566 and parameters: {'classifier': 'SVM', 'C': 0.35514221066922197, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,871] Trial 81 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.13626758922790153, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,910] Trial 82 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.1762270732558732, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,947] Trial 83 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.12456553676912333, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:51,985] Trial 84 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.24080138501900222, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,024] Trial 85 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.12868220784880763, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,058] Trial 86 finished with value: 0.7858472998137803 and

parameters: {'classifier': 'SVM', 'C': 0.19156830346290685, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,694] Trial 87 finished with value: 0.7746741154562384 and parameters: {'classifier': 'RandomForest', 'n_estimators': 99, 'max_depth': 11, 'min_samples_split': 3, 'min_samples_leaf': 8, 'bootstrap': False}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,750] Trial 88 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 4.272388819334623, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,790] Trial 89 finished with value: 0.7616387337057727 and parameters: {'classifier': 'SVM', 'C': 0.2820856994765886, 'kernel': 'sigmoid', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,832] Trial 90 finished with value: 0.7150837988826816 and parameters: {'classifier': 'SVM', 'C': 0.15468496397739726, 'kernel': 'poly', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,880] Trial 91 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.12856589550782094, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,919] Trial 92 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.16481033824014227, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:52,960] Trial 93 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.1308562879679902, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:53,029] Trial 94 finished with value: 0.7858472998137801 and parameters: {'classifier': 'SVM', 'C': 7.6498994521629795, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:53,068] Trial 95 finished with value: 0.7858472998137803 and parameters: {'classifier': 'SVM', 'C': 0.21188772036390022, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:53,109] Trial 96 finished with value: 0.7895716945996275 and parameters: {'classifier': 'SVM', 'C': 0.1226840265479702, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:54,708] Trial 97 finished with value: 0.7374301675977654 and parameters: {'classifier': 'GradientBoosting', 'n_estimators': 51, 'learning_rate': 0.011403197087115572, 'max_depth': 8, 'min_samples_split': 7, 'min_samples_leaf': 3}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:54,809] Trial 98 finished with value: 0.7541899441340782 and parameters: {'classifier': 'SVM', 'C': 0.15589688419129144, 'kernel': 'rbf', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

[I 2025-12-06 14:54:54,863] Trial 99 finished with value: 0.7877094972067038 and parameters: {'classifier': 'SVM', 'C': 0.10016799925753282, 'kernel': 'linear', 'gamma': 'auto'}. Best is trial 42 with value: 0.7895716945996275.

```
[32]: # Retrieve the best trial
      best_trial = study.best_trial
      print("Best trial parameters:", best_trial.params)
```

```
print("Best trial accuracy:", best_trial.value)
```

```
Best trial parameters: {'classifier': 'SVM', 'C': 0.11063344610550901, 'kernel': 'linear', 'gamma': 'scale'}
```

```
Best trial accuracy: 0.7895716945996275
```

```
[33]: study.trials_dataframe()
```

```
[33]:
```

	number	value	datetime_start	datetime_complete	\
0	0	0.769088	2025-12-06 14:53:59.948435	2025-12-06 14:54:03.194215	
1	1	0.770950	2025-12-06 14:54:03.196223	2025-12-06 14:54:03.257176	
2	2	0.756052	2025-12-06 14:54:03.259213	2025-12-06 14:54:06.632967	
3	3	0.718808	2025-12-06 14:54:06.647498	2025-12-06 14:54:06.727952	
4	4	0.785847	2025-12-06 14:54:06.730888	2025-12-06 14:54:06.920162	
..	...	...	...	...	
95	95	0.785847	2025-12-06 14:54:53.029659	2025-12-06 14:54:53.068754	
96	96	0.789572	2025-12-06 14:54:53.068754	2025-12-06 14:54:53.109189	
97	97	0.737430	2025-12-06 14:54:53.109189	2025-12-06 14:54:54.708466	
98	98	0.754190	2025-12-06 14:54:54.708466	2025-12-06 14:54:54.809550	
99	99	0.787709	2025-12-06 14:54:54.809550	2025-12-06 14:54:54.863576	

	duration	params_C	params_bootstrap	params_classifier	\
0	0 days 00:00:03.245780	NaN	True	RandomForest	
1	0 days 00:00:00.060953	1.567108	NaN	SVM	
2	0 days 00:00:03.373754	NaN	True	RandomForest	
3	0 days 00:00:00.080454	44.811488	NaN	SVM	
4	0 days 00:00:00.189274	23.772037	NaN	SVM	
..	...	...	...	...	
95	0 days 00:00:00.039095	0.211888	NaN	SVM	
96	0 days 00:00:00.040435	0.122684	NaN	SVM	
97	0 days 00:00:01.599277	NaN	NaN	GradientBoosting	
98	0 days 00:00:00.101084	0.155897	NaN	SVM	
99	0 days 00:00:00.054026	0.100168	NaN	SVM	

	params_gamma	params_kernel	params_learning_rate	params_max_depth	\
0	NaN	NaN	NaN	10.0	
1	auto	rbf	NaN	NaN	
2	NaN	NaN	NaN	9.0	
3	auto	rbf	NaN	NaN	
4	auto	linear	NaN	NaN	
..	...	...	...	...	
95	auto	linear	NaN	NaN	
96	auto	linear	NaN	NaN	
97	NaN	NaN	0.011403	8.0	
98	auto	rbf	NaN	NaN	
99	auto	linear	NaN	NaN	

	params_min_samples_leaf	params_min_samples_split	params_n_estimators	\
0	3.0	4.0	174.0	
1	NaN	NaN	NaN	
2	10.0	5.0	270.0	
3	NaN	NaN	NaN	
4	NaN	NaN	NaN	
..	...	...	...	
95	NaN	NaN	NaN	
96	NaN	NaN	NaN	
97	3.0	7.0	51.0	
98	NaN	NaN	NaN	
99	NaN	NaN	NaN	

	state
0	COMPLETE
1	COMPLETE
2	COMPLETE
3	COMPLETE
4	COMPLETE
..	...
95	COMPLETE
96	COMPLETE
97	COMPLETE
98	COMPLETE
99	COMPLETE

[100 rows x 16 columns]

```
[34]: study.trials_dataframe()['params_classifier'].value_counts()
```

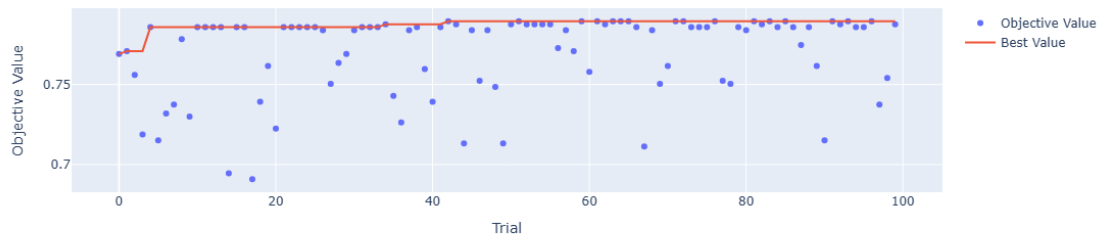
```
[34]: params_classifier
SVM                80
RandomForest       10
GradientBoosting    10
Name: count, dtype: int64
```

```
[35]: study.trials_dataframe().groupby('params_classifier')['value'].mean()
```

```
[35]: params_classifier
GradientBoosting    0.744134
RandomForest        0.767039
SVM                 0.774278
Name: value, dtype: float64
```

```
[36]: # 1. Optimization History
plot_optimization_history(study).show()
```

Optimization History Plot



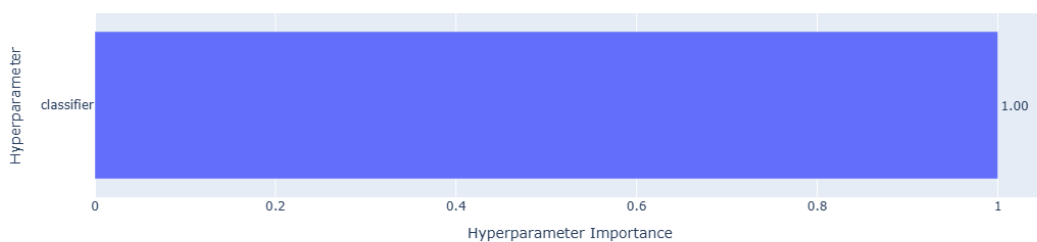
```
[37]: # 3. Slice Plot
plot_slice(study).show()
```

Slice Plot



```
[38]: # 5. Hyperparameter Importance
plot_param_importances(study).show()
```

Hyperparameter Importances




```

[39]: import optuna
import xgboost as xgb
from sklearn.model_selection import train_test_split
from sklearn.datasets import load_iris
from sklearn.metrics import accuracy_score
import numpy as np

# Load the Iris dataset
X, y = load_iris(return_X_y=True)

# Split the dataset into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    ↪random_state=42)

# Define the objective function for XGBoost
def objective(trial):
    # Hyperparameter search space
    param = {
        'verbosity': 0,
        'objective': 'multi:softprob',
        'num_class': 3,
        'eval_metric': 'mlogloss', # Ensure that the eval_metric is specified
    ↪here
        'booster': 'gbtree',
        'lambda': trial.suggest_float('lambda', 1e-8, 1.0, log=True),
        'alpha': trial.suggest_float('alpha', 1e-8, 1.0, log=True),
        'eta': trial.suggest_float('eta', 0.01, 0.3),
        'gamma': trial.suggest_float('gamma', 1e-8, 1.0, log=True),
        'max_depth': trial.suggest_int('max_depth', 3, 9),
        'min_child_weight': trial.suggest_int('min_child_weight', 1, 10),
        'subsample': trial.suggest_float('subsample', 0.4, 1.0),
        'colsample_bytree': trial.suggest_float('colsample_bytree', 0.4, 1.0),
        'n_estimators': 300,
    }

    # Create DMatrix for XGBoost
    dtrain = xgb.DMatrix(X_train, label=y_train)
    dtest = xgb.DMatrix(X_test, label=y_test)

    # Define a pruning callback based on evaluation metrics
    pruning_callback = optuna.integration.XGBoostPruningCallback(trial,
    ↪"eval-mlogloss") # Match the metric name in the evals list

    # Train the model
    bst = xgb.train(
        param,
        dtrain,

```

```

        num_boost_round=300,
        evals=[(dtrain, "train"), (dtest, "eval")], # Ensure the eval datasets
↳and names are specified
        early_stopping_rounds=30,
        callbacks=[pruning_callback]
    )

    # Predict on the test set
    preds = bst.predict(dtest)
    best_preds = [int(np.argmax(line)) for line in preds]

    # Return accuracy as the objective value
    accuracy = accuracy_score(y_test, best_preds)
    return accuracy

# Create a study with pruning
study = optuna.create_study(direction='maximize', pruner=optuna.pruners.
↳SuccessiveHalvingPruner())
study.optimize(objective, n_trials=50)

# Output the best trial
print(f"Best trial: {study.best_trial.params}")
print(f"Best accuracy: {study.best_value}")

```

[I 2025-12-06 14:55:10,949] A new study created in memory with name: no-name-b778bfc1-d10b-4561-892c-9d4e4483b5d9

```

[0]    train-mlogloss:0.90915  eval-mlogloss:0.90225
[1]    train-mlogloss:0.84249  eval-mlogloss:0.81771
[2]    train-mlogloss:0.78066  eval-mlogloss:0.75396
[3]    train-mlogloss:0.66606  eval-mlogloss:0.63104
[4]    train-mlogloss:0.60149  eval-mlogloss:0.56177
[5]    train-mlogloss:0.53998  eval-mlogloss:0.49313
[6]    train-mlogloss:0.49173  eval-mlogloss:0.43978
[7]    train-mlogloss:0.45717  eval-mlogloss:0.40350
[8]    train-mlogloss:0.43602  eval-mlogloss:0.38390
[9]    train-mlogloss:0.40409  eval-mlogloss:0.35087
[10]   train-mlogloss:0.39035  eval-mlogloss:0.33423
[11]   train-mlogloss:0.38740  eval-mlogloss:0.33056
[12]   train-mlogloss:0.37430  eval-mlogloss:0.31765
[13]   train-mlogloss:0.37398  eval-mlogloss:0.31748
[14]   train-mlogloss:0.37291  eval-mlogloss:0.31553
[15]   train-mlogloss:0.37264  eval-mlogloss:0.31490
[16]   train-mlogloss:0.37160  eval-mlogloss:0.31346
[17]   train-mlogloss:0.37020  eval-mlogloss:0.31206
[18]   train-mlogloss:0.36696  eval-mlogloss:0.30626
[19]   train-mlogloss:0.36430  eval-mlogloss:0.30278
[20]   train-mlogloss:0.36438  eval-mlogloss:0.30256

```

[21]	train-mlogloss:0.36425	eval-mlogloss:0.30278
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[I 2025-12-06 14:55:11,839] Trial 0 finished with value: 1.0 and parameters: {'lambda': 0.03294147801498702, 'alpha': 0.0007672671426894849, 'eta': 0.15847595071800358, 'gamma': 1.0731954455671145e-08, 'max_depth': 4, 'min_child_weight': 8, 'subsample': 0.46617547356234623, 'colsample_bytree': 0.6868533439045852}. Best is trial 0 with value: 1.0.

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[120]	train-mlogloss:0.18109	eval-mlogloss:0.11969
[121]	train-mlogloss:0.18106	eval-mlogloss:0.11986
[122]	train-mlogloss:0.18106	eval-mlogloss:0.12021
[123]	train-mlogloss:0.18095	eval-mlogloss:0.12050
[124]	train-mlogloss:0.18092	eval-mlogloss:0.11984
[125]	train-mlogloss:0.18091	eval-mlogloss:0.11976
[126]	train-mlogloss:0.18091	eval-mlogloss:0.11990
[127]	train-mlogloss:0.18079	eval-mlogloss:0.11969
[128]	train-mlogloss:0.18071	eval-mlogloss:0.11983
[129]	train-mlogloss:0.18067	eval-mlogloss:0.11983
[130]	train-mlogloss:0.18054	eval-mlogloss:0.11945
[131]	train-mlogloss:0.18055	eval-mlogloss:0.11905
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[133]	train-mlogloss:0.18051	eval-mlogloss:0.11942
[134]	train-mlogloss:0.18051	eval-mlogloss:0.11946
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[138]	train-mlogloss:0.18025	eval-mlogloss:0.11924
[139]	train-mlogloss:0.18012	eval-mlogloss:0.11895
[140]	train-mlogloss:0.18008	eval-mlogloss:0.11898
[141]	train-mlogloss:0.17988	eval-mlogloss:0.11893
[142]	train-mlogloss:0.17989	eval-mlogloss:0.11899
[143]	train-mlogloss:0.17987	eval-mlogloss:0.11890
[144]	train-mlogloss:0.17993	eval-mlogloss:0.11908
[145]	train-mlogloss:0.17987	eval-mlogloss:0.11889
[146]	train-mlogloss:0.17959	eval-mlogloss:0.11903
[147]	train-mlogloss:0.17954	eval-mlogloss:0.11894
[148]	train-mlogloss:0.17955	eval-mlogloss:0.11902
[149]	train-mlogloss:0.17949	eval-mlogloss:0.11883
[150]	train-mlogloss:0.17946	eval-mlogloss:0.11864
[151]	train-mlogloss:0.17947	eval-mlogloss:0.11838
[152]	train-mlogloss:0.17941	eval-mlogloss:0.11831

[153]	train-mlogloss:0.17943	eval-mlogloss:0.11831
[154]	train-mlogloss:0.17939	eval-mlogloss:0.11826
[155]	train-mlogloss:0.17933	eval-mlogloss:0.11864
[156]	train-mlogloss:0.17933	eval-mlogloss:0.11863
[157]	train-mlogloss:0.17935	eval-mlogloss:0.11871
[158]	train-mlogloss:0.17934	eval-mlogloss:0.11853
[159]	train-mlogloss:0.17929	eval-mlogloss:0.11841
[160]	train-mlogloss:0.17929	eval-mlogloss:0.11856
[161]	train-mlogloss:0.17929	eval-mlogloss:0.11842
[162]	train-mlogloss:0.17924	eval-mlogloss:0.11845
[163]	train-mlogloss:0.17928	eval-mlogloss:0.11855
[164]	train-mlogloss:0.17943	eval-mlogloss:0.11856
[165]	train-mlogloss:0.17929	eval-mlogloss:0.11815
[166]	train-mlogloss:0.17929	eval-mlogloss:0.11864
[167]	train-mlogloss:0.17925	eval-mlogloss:0.11852
[168]	train-mlogloss:0.17925	eval-mlogloss:0.11841
[169]	train-mlogloss:0.17922	eval-mlogloss:0.11826
[170]	train-mlogloss:0.17919	eval-mlogloss:0.11832
[171]	train-mlogloss:0.17918	eval-mlogloss:0.11835
[172]	train-mlogloss:0.17921	eval-mlogloss:0.11868
[173]	train-mlogloss:0.17918	eval-mlogloss:0.11881
[174]	train-mlogloss:0.17917	eval-mlogloss:0.11906
[175]	train-mlogloss:0.17913	eval-mlogloss:0.11903
[176]	train-mlogloss:0.17917	eval-mlogloss:0.11914
[177]	train-mlogloss:0.17898	eval-mlogloss:0.11930
[178]	train-mlogloss:0.17894	eval-mlogloss:0.11958
[179]	train-mlogloss:0.17895	eval-mlogloss:0.11950
[180]	train-mlogloss:0.17897	eval-mlogloss:0.11859
[181]	train-mlogloss:0.17896	eval-mlogloss:0.11874
[182]	train-mlogloss:0.17891	eval-mlogloss:0.11870
[183]	train-mlogloss:0.17888	eval-mlogloss:0.11882
[184]	train-mlogloss:0.17882	eval-mlogloss:0.11882
[185]	train-mlogloss:0.17882	eval-mlogloss:0.11883
[186]	train-mlogloss:0.17877	eval-mlogloss:0.11867
[187]	train-mlogloss:0.17874	eval-mlogloss:0.11864
[188]	train-mlogloss:0.17876	eval-mlogloss:0.11859
[189]	train-mlogloss:0.17877	eval-mlogloss:0.11863
[190]	train-mlogloss:0.17876	eval-mlogloss:0.11883
[191]	train-mlogloss:0.17875	eval-mlogloss:0.11875
[192]	train-mlogloss:0.17875	eval-mlogloss:0.11875
[193]	train-mlogloss:0.17876	eval-mlogloss:0.11870
[194]	train-mlogloss:0.17878	eval-mlogloss:0.11888
[195]	train-mlogloss:0.17870	eval-mlogloss:0.11879

[I 2025-12-06 14:55:12,599] Trial 1 finished with value: 1.0 and parameters: {'lambda': 1.231542259888964e-07, 'alpha': 1.4195081324725755e-07, 'eta': 0.260006533930035, 'gamma': 0.011028253334656998, 'max_depth': 9, 'min_child_weight': 8, 'subsample': 0.8751697751759812, 'colsample_bytree':

0.6806450451295118}. Best is trial 0 with value: 1.0.

[0]	train-mlogloss:0.82645	eval-mlogloss:0.81004
[1]	train-mlogloss:0.72447	eval-mlogloss:0.70395
[2]	train-mlogloss:0.58608	eval-mlogloss:0.55590
[3]	train-mlogloss:0.46754	eval-mlogloss:0.43480
[4]	train-mlogloss:0.38580	eval-mlogloss:0.35286
[5]	train-mlogloss:0.32042	eval-mlogloss:0.28065
[6]	train-mlogloss:0.27701	eval-mlogloss:0.22942
[7]	train-mlogloss:0.24117	eval-mlogloss:0.18815
[8]	train-mlogloss:0.21625	eval-mlogloss:0.15726
[9]	train-mlogloss:0.19912	eval-mlogloss:0.13762
[10]	train-mlogloss:0.19359	eval-mlogloss:0.13077
[11]	train-mlogloss:0.18737	eval-mlogloss:0.12416
[12]	train-mlogloss:0.18695	eval-mlogloss:0.12362
[13]	train-mlogloss:0.18198	eval-mlogloss:0.11689
[14]	train-mlogloss:0.18174	eval-mlogloss:0.11636
[15]	train-mlogloss:0.18168	eval-mlogloss:0.11614

[I 2025-12-06 14:55:12,665] Trial 2 pruned. Trial was pruned at iteration 16.

[0]	train-mlogloss:1.02491	eval-mlogloss:1.02645
[1]	train-mlogloss:0.98007	eval-mlogloss:0.97244
[2]	train-mlogloss:0.93648	eval-mlogloss:0.92808
[3]	train-mlogloss:0.87332	eval-mlogloss:0.86186
[4]	train-mlogloss:0.82250	eval-mlogloss:0.80924
[5]	train-mlogloss:0.77542	eval-mlogloss:0.75844
[6]	train-mlogloss:0.74170	eval-mlogloss:0.72018
[7]	train-mlogloss:0.70955	eval-mlogloss:0.68654
[8]	train-mlogloss:0.68013	eval-mlogloss:0.65439
[9]	train-mlogloss:0.65285	eval-mlogloss:0.62421
[10]	train-mlogloss:0.62684	eval-mlogloss:0.59446
[11]	train-mlogloss:0.59572	eval-mlogloss:0.56170
[12]	train-mlogloss:0.57110	eval-mlogloss:0.53547
[13]	train-mlogloss:0.54986	eval-mlogloss:0.51319
[14]	train-mlogloss:0.53226	eval-mlogloss:0.49472
[15]	train-mlogloss:0.51224	eval-mlogloss:0.47326
[16]	train-mlogloss:0.49669	eval-mlogloss:0.45721
[17]	train-mlogloss:0.48240	eval-mlogloss:0.44159
[18]	train-mlogloss:0.46764	eval-mlogloss:0.42630
[19]	train-mlogloss:0.45139	eval-mlogloss:0.40864
[20]	train-mlogloss:0.43801	eval-mlogloss:0.39339
[21]	train-mlogloss:0.42553	eval-mlogloss:0.37855
[22]	train-mlogloss:0.41897	eval-mlogloss:0.37094
[23]	train-mlogloss:0.40681	eval-mlogloss:0.35739
[24]	train-mlogloss:0.39563	eval-mlogloss:0.34482
[25]	train-mlogloss:0.38582	eval-mlogloss:0.33384
[26]	train-mlogloss:0.38148	eval-mlogloss:0.32887
[27]	train-mlogloss:0.37686	eval-mlogloss:0.32388

[28]	train-mlogloss:0.37667	eval-mlogloss:0.32380
[29]	train-mlogloss:0.37498	eval-mlogloss:0.32191
[30]	train-mlogloss:0.37046	eval-mlogloss:0.31747
[31]	train-mlogloss:0.36529	eval-mlogloss:0.31214
[32]	train-mlogloss:0.36442	eval-mlogloss:0.31134
[33]	train-mlogloss:0.36417	eval-mlogloss:0.31141
[34]	train-mlogloss:0.35947	eval-mlogloss:0.30573
[35]	train-mlogloss:0.35919	eval-mlogloss:0.30580
[36]	train-mlogloss:0.35777	eval-mlogloss:0.30413
[37]	train-mlogloss:0.35368	eval-mlogloss:0.30001
[38]	train-mlogloss:0.34881	eval-mlogloss:0.29533
[39]	train-mlogloss:0.34864	eval-mlogloss:0.29528
[40]	train-mlogloss:0.34797	eval-mlogloss:0.29432
[41]	train-mlogloss:0.34750	eval-mlogloss:0.29391
[42]	train-mlogloss:0.34739	eval-mlogloss:0.29385
[43]	train-mlogloss:0.34716	eval-mlogloss:0.29336
[44]	train-mlogloss:0.34709	eval-mlogloss:0.29330
[45]	train-mlogloss:0.34693	eval-mlogloss:0.29315
[46]	train-mlogloss:0.34670	eval-mlogloss:0.29221
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[48]	train-mlogloss:0.34584	eval-mlogloss:0.29015
[49]	train-mlogloss:0.34507	eval-mlogloss:0.28931
[50]	train-mlogloss:0.34089	eval-mlogloss:0.28427
[51]	train-mlogloss:0.33687	eval-mlogloss:0.27796
[52]	train-mlogloss:0.33658	eval-mlogloss:0.27756
[53]	train-mlogloss:0.33645	eval-mlogloss:0.27746
[54]	train-mlogloss:0.33628	eval-mlogloss:0.27716
[55]	train-mlogloss:0.33614	eval-mlogloss:0.27693
[56]	train-mlogloss:0.33586	eval-mlogloss:0.27647
[57]	train-mlogloss:0.33557	eval-mlogloss:0.27540
[58]	train-mlogloss:0.33558	eval-mlogloss:0.27543
[59]	train-mlogloss:0.33531	eval-mlogloss:0.27546
[60]	train-mlogloss:0.33501	eval-mlogloss:0.27489
[61]	train-mlogloss:0.33463	eval-mlogloss:0.27471
[62]	train-mlogloss:0.33442	eval-mlogloss:0.27454
[63]	train-mlogloss:0.33443	eval-mlogloss:0.27458
[64]	train-mlogloss:0.33417	eval-mlogloss:0.27465
[65]	train-mlogloss:0.33422	eval-mlogloss:0.27485
[66]	train-mlogloss:0.33369	eval-mlogloss:0.27440
[67]	train-mlogloss:0.33351	eval-mlogloss:0.27444
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[70]	train-mlogloss:0.33261	eval-mlogloss:0.27371
[71]	train-mlogloss:0.33231	eval-mlogloss:0.27331
[72]	train-mlogloss:0.33232	eval-mlogloss:0.27322
[73]	train-mlogloss:0.33206	eval-mlogloss:0.27304
[74]	train-mlogloss:0.33171	eval-mlogloss:0.27333
[75]	train-mlogloss:0.33136	eval-mlogloss:0.27324

[76]	train-mlogloss:0.33085	eval-mlogloss:0.27260
[77]	train-mlogloss:0.33052	eval-mlogloss:0.27244
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[83]	train-mlogloss:0.33013	eval-mlogloss:0.27210
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[91]	train-mlogloss:0.32672	eval-mlogloss:0.26892
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[94]	train-mlogloss:0.32622	eval-mlogloss:0.26802
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[103]	train-mlogloss:0.32347	eval-mlogloss:0.26532
[104]	train-mlogloss:0.32317	eval-mlogloss:0.26527
[105]	train-mlogloss:0.32304	eval-mlogloss:0.26514
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[115]	train-mlogloss:0.32150	eval-mlogloss:0.26224
[116]	train-mlogloss:0.32130	eval-mlogloss:0.26233
[117]	train-mlogloss:0.32097	eval-mlogloss:0.26163
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[120]	train-mlogloss:0.32038	eval-mlogloss:0.26105
[121]	train-mlogloss:0.32016	eval-mlogloss:0.26103
[122]	train-mlogloss:0.32014	eval-mlogloss:0.26108
[123]	train-mlogloss:0.32010	eval-mlogloss:0.26087

[124]	train-mlogloss:0.31664	eval-mlogloss:0.25704
[125]	train-mlogloss:0.31651	eval-mlogloss:0.25717
[126]	train-mlogloss:0.31627	eval-mlogloss:0.25722
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[130]	train-mlogloss:0.31587	eval-mlogloss:0.25712
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[137]	train-mlogloss:0.31494	eval-mlogloss:0.25583
[138]	train-mlogloss:0.31475	eval-mlogloss:0.25544
[139]	train-mlogloss:0.31454	eval-mlogloss:0.25535
[140]	train-mlogloss:0.31439	eval-mlogloss:0.25518
[141]	train-mlogloss:0.31419	eval-mlogloss:0.25446
[142]	train-mlogloss:0.31425	eval-mlogloss:0.25459
[143]	train-mlogloss:0.31419	eval-mlogloss:0.25464
[144]	train-mlogloss:0.31391	eval-mlogloss:0.25373
[145]	train-mlogloss:0.31382	eval-mlogloss:0.25383
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[150]	train-mlogloss:0.31350	eval-mlogloss:0.25269
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[152]	train-mlogloss:0.31354	eval-mlogloss:0.25270
[153]	train-mlogloss:0.31342	eval-mlogloss:0.25268
[154]	train-mlogloss:0.31345	eval-mlogloss:0.25238
[155]	train-mlogloss:0.31351	eval-mlogloss:0.25240
[156]	train-mlogloss:0.31355	eval-mlogloss:0.25247
[157]	train-mlogloss:0.31353	eval-mlogloss:0.25235
[158]	train-mlogloss:0.31351	eval-mlogloss:0.25237
[159]	train-mlogloss:0.31345	eval-mlogloss:0.25236
[160]	train-mlogloss:0.31343	eval-mlogloss:0.25224
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[164]	train-mlogloss:0.31297	eval-mlogloss:0.25220
[165]	train-mlogloss:0.31283	eval-mlogloss:0.25156
[166]	train-mlogloss:0.31277	eval-mlogloss:0.25121
[167]	train-mlogloss:0.31269	eval-mlogloss:0.25136
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[191]	train-mlogloss:0.31118	eval-mlogloss:0.24951
[192]	train-mlogloss:0.31116	eval-mlogloss:0.24958
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[194]	train-mlogloss:0.31089	eval-mlogloss:0.24908
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[202]	train-mlogloss:0.31013	eval-mlogloss:0.24749
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[212]	train-mlogloss:0.30987	eval-mlogloss:0.24613
[213]	train-mlogloss:0.30979	eval-mlogloss:0.24613
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[220]	train-mlogloss:0.30972	eval-mlogloss:0.24567
[221]	train-mlogloss:0.30969	eval-mlogloss:0.24582
[222]	train-mlogloss:0.30969	eval-mlogloss:0.24582
[223]	train-mlogloss:0.30968	eval-mlogloss:0.24588
[224]	train-mlogloss:0.30958	eval-mlogloss:0.24595
[225]	train-mlogloss:0.30933	eval-mlogloss:0.24574
[226]	train-mlogloss:0.30936	eval-mlogloss:0.24552
[227]	train-mlogloss:0.30936	eval-mlogloss:0.24560
[228]	train-mlogloss:0.30928	eval-mlogloss:0.24571
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[254]	train-mlogloss:0.30835	eval-mlogloss:0.24423
[255]	train-mlogloss:0.30826	eval-mlogloss:0.24436
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[260]	train-mlogloss:0.30831	eval-mlogloss:0.24404
[261]	train-mlogloss:0.30825	eval-mlogloss:0.24403
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[263]	train-mlogloss:0.30805	eval-mlogloss:0.24321
[264]	train-mlogloss:0.30811	eval-mlogloss:0.24323
[265]	train-mlogloss:0.30811	eval-mlogloss:0.24325
[266]	train-mlogloss:0.30804	eval-mlogloss:0.24315
[267]	train-mlogloss:0.30787	eval-mlogloss:0.24304


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[268] train-mlogloss:0.30788 eval-mlogloss:0.24300
[269] train-mlogloss:0.30783 eval-mlogloss:0.24305
[270] train-mlogloss:0.30785 eval-mlogloss:0.24292
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[278] train-mlogloss:0.30774 eval-mlogloss:0.24311
[279] train-mlogloss:0.30775 eval-mlogloss:0.24305
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[291] train-mlogloss:0.30720 eval-mlogloss:0.24304
[292] train-mlogloss:0.30720 eval-mlogloss:0.24306
[293] train-mlogloss:0.30696 eval-mlogloss:0.24294
[294] train-mlogloss:0.30696 eval-mlogloss:0.24298
[295] train-mlogloss:0.30695 eval-mlogloss:0.24284
[296] train-mlogloss:0.30695 eval-mlogloss:0.24302
[297] train-mlogloss:0.30696 eval-mlogloss:0.24318
[298] train-mlogloss:0.30689 eval-mlogloss:0.24319
[299] train-mlogloss:0.30688 eval-mlogloss:0.24306

```

[I 2025-12-06 14:55:13,796] Trial 3 finished with value: 0.9666666666666667 and parameters: {'lambda': 2.6711003450264135e-07, 'alpha': 0.001370213463465561, 'eta': 0.06000950043066181, 'gamma': 0.00027623986442692415, 'max_depth': 4, 'min_child_weight': 9, 'subsample': 0.5467746319469173, 'colsample_bytree': 0.87530051129099}. Best is trial 0 with value: 1.0.

```

[0] train-mlogloss:0.87198 eval-mlogloss:0.86166

```

[I 2025-12-06 14:55:13,811] Trial 4 pruned. Trial was pruned at iteration 1.

```

[0] train-mlogloss:0.77561 eval-mlogloss:0.75967

```

[I 2025-12-06 14:55:13,824] Trial 5 pruned. Trial was pruned at iteration 1.

```

[0] train-mlogloss:0.90057 eval-mlogloss:0.88875

```

[I 2025-12-06 14:55:13,838] Trial 6 pruned. Trial was pruned at iteration 1.

```

[0] train-mlogloss:0.84471 eval-mlogloss:0.83858

```

[I 2025-12-06 14:55:13,851] Trial 7 pruned. Trial was pruned at iteration 1.

[0] train-mlogloss:0.88250 eval-mlogloss:0.87186

[I 2025-12-06 14:55:13,864] Trial 8 pruned. Trial was pruned at iteration 1.

[0] train-mlogloss:0.79715 eval-mlogloss:0.80176

[I 2025-12-06 14:55:13,877] Trial 9 pruned. Trial was pruned at iteration 1.

[0] train-mlogloss:1.03888 eval-mlogloss:1.03710

[1] train-mlogloss:0.99643 eval-mlogloss:0.98966

[2] train-mlogloss:0.94897 eval-mlogloss:0.94002

[3] train-mlogloss:0.90550 eval-mlogloss:0.89348

[4] train-mlogloss:0.85510 eval-mlogloss:0.84258

[5] train-mlogloss:0.82357 eval-mlogloss:0.80652

[6] train-mlogloss:0.78462 eval-mlogloss:0.76665

[7] train-mlogloss:0.76517 eval-mlogloss:0.75009

[8] train-mlogloss:0.74225 eval-mlogloss:0.72344

[9] train-mlogloss:0.70811 eval-mlogloss:0.68810

[10] train-mlogloss:0.69316 eval-mlogloss:0.67317

[11] train-mlogloss:0.67532 eval-mlogloss:0.64884

[12] train-mlogloss:0.65579 eval-mlogloss:0.63125

[13] train-mlogloss:0.65441 eval-mlogloss:0.63018

[14] train-mlogloss:0.65155 eval-mlogloss:0.62524

[15] train-mlogloss:0.62400 eval-mlogloss:0.59559

[16] train-mlogloss:0.62208 eval-mlogloss:0.59381

[17] train-mlogloss:0.62033 eval-mlogloss:0.59228

[18] train-mlogloss:0.61489 eval-mlogloss:0.58490

[19] train-mlogloss:0.61289 eval-mlogloss:0.58321

[20] train-mlogloss:0.60639 eval-mlogloss:0.57906

[21] train-mlogloss:0.59257 eval-mlogloss:0.56528

[22] train-mlogloss:0.58980 eval-mlogloss:0.55977

[23] train-mlogloss:0.56352 eval-mlogloss:0.53205

[24] train-mlogloss:0.55879 eval-mlogloss:0.52691

[25] train-mlogloss:0.54567 eval-mlogloss:0.51294

[26] train-mlogloss:0.54420 eval-mlogloss:0.51209

[27] train-mlogloss:0.54425 eval-mlogloss:0.51228

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[29] train-mlogloss:0.54261 eval-mlogloss:0.50909

[30] train-mlogloss:0.54175 eval-mlogloss:0.50903

[31] train-mlogloss:0.53943 eval-mlogloss:0.50656

[32] train-mlogloss:0.53908 eval-mlogloss:0.50670

[33] train-mlogloss:0.53904 eval-mlogloss:0.50604

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[35] train-mlogloss:0.53911 eval-mlogloss:0.50589

[36] train-mlogloss:0.53359 eval-mlogloss:0.50059

[37] train-mlogloss:0.52160 eval-mlogloss:0.48807

[38] train-mlogloss:0.51522 eval-mlogloss:0.48183

[39] train-mlogloss:0.51245 eval-mlogloss:0.47847

[40]	train-mlogloss:0.51135	eval-mlogloss:0.47790
[41]	train-mlogloss:0.51031	eval-mlogloss:0.47646
[42]	train-mlogloss:0.51011	eval-mlogloss:0.47642
[43]	train-mlogloss:0.51004	eval-mlogloss:0.47664
[44]	train-mlogloss:0.50997	eval-mlogloss:0.47628
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[187]	train-mlogloss:0.44578	eval-mlogloss:0.39704
[188]	train-mlogloss:0.44572	eval-mlogloss:0.39715
[189]	train-mlogloss:0.44567	eval-mlogloss:0.39762
[190]	train-mlogloss:0.44579	eval-mlogloss:0.39785
[191]	train-mlogloss:0.44582	eval-mlogloss:0.39777
[192]	train-mlogloss:0.44580	eval-mlogloss:0.39800
[193]	train-mlogloss:0.44580	eval-mlogloss:0.39765
[194]	train-mlogloss:0.44560	eval-mlogloss:0.39773
[195]	train-mlogloss:0.44565	eval-mlogloss:0.39742
[196]	train-mlogloss:0.44568	eval-mlogloss:0.39771
[197]	train-mlogloss:0.44563	eval-mlogloss:0.39736
[198]	train-mlogloss:0.44574	eval-mlogloss:0.39683
[199]	train-mlogloss:0.44529	eval-mlogloss:0.39639
[200]	train-mlogloss:0.44527	eval-mlogloss:0.39653
[201]	train-mlogloss:0.44539	eval-mlogloss:0.39618
[202]	train-mlogloss:0.44537	eval-mlogloss:0.39635
[203]	train-mlogloss:0.44530	eval-mlogloss:0.39671
[204]	train-mlogloss:0.44528	eval-mlogloss:0.39681
[205]	train-mlogloss:0.44535	eval-mlogloss:0.39666
[206]	train-mlogloss:0.44506	eval-mlogloss:0.39603
[207]	train-mlogloss:0.44508	eval-mlogloss:0.39622
[208]	train-mlogloss:0.44492	eval-mlogloss:0.39642
[209]	train-mlogloss:0.44503	eval-mlogloss:0.39601
[210]	train-mlogloss:0.44501	eval-mlogloss:0.39614
[211]	train-mlogloss:0.44491	eval-mlogloss:0.39592
[212]	train-mlogloss:0.44491	eval-mlogloss:0.39604
[213]	train-mlogloss:0.44428	eval-mlogloss:0.39562
[214]	train-mlogloss:0.44438	eval-mlogloss:0.39518
[215]	train-mlogloss:0.44349	eval-mlogloss:0.39503
[216]	train-mlogloss:0.44349	eval-mlogloss:0.39500
[217]	train-mlogloss:0.44318	eval-mlogloss:0.39422
[218]	train-mlogloss:0.44281	eval-mlogloss:0.39347
[219]	train-mlogloss:0.44287	eval-mlogloss:0.39299
[220]	train-mlogloss:0.44287	eval-mlogloss:0.39287
[221]	train-mlogloss:0.44280	eval-mlogloss:0.39314
[222]	train-mlogloss:0.44280	eval-mlogloss:0.39331
[223]	train-mlogloss:0.44276	eval-mlogloss:0.39341
[224]	train-mlogloss:0.44250	eval-mlogloss:0.39333
[225]	train-mlogloss:0.44250	eval-mlogloss:0.39350
[226]	train-mlogloss:0.44251	eval-mlogloss:0.39295
[227]	train-mlogloss:0.44250	eval-mlogloss:0.39317
[228]	train-mlogloss:0.44249	eval-mlogloss:0.39341
[229]	train-mlogloss:0.44255	eval-mlogloss:0.39401
[230]	train-mlogloss:0.44188	eval-mlogloss:0.39153
[231]	train-mlogloss:0.44203	eval-mlogloss:0.39201

[232]	train-mlogloss:0.44195	eval-mlogloss:0.39201
[233]	train-mlogloss:0.44193	eval-mlogloss:0.39171
[234]	train-mlogloss:0.44204	eval-mlogloss:0.39217
[235]	train-mlogloss:0.44206	eval-mlogloss:0.39247
[236]	train-mlogloss:0.44203	eval-mlogloss:0.39230
[237]	train-mlogloss:0.44204	eval-mlogloss:0.39225
[238]	train-mlogloss:0.44202	eval-mlogloss:0.39209
[239]	train-mlogloss:0.44182	eval-mlogloss:0.39192
[240]	train-mlogloss:0.44098	eval-mlogloss:0.38873
[241]	train-mlogloss:0.44086	eval-mlogloss:0.38806
[242]	train-mlogloss:0.44101	eval-mlogloss:0.38865
[243]	train-mlogloss:0.44095	eval-mlogloss:0.38835
[244]	train-mlogloss:0.44100	eval-mlogloss:0.38839
[245]	train-mlogloss:0.44096	eval-mlogloss:0.38838
[246]	train-mlogloss:0.44084	eval-mlogloss:0.38718
[247]	train-mlogloss:0.44086	eval-mlogloss:0.38734
[248]	train-mlogloss:0.44092	eval-mlogloss:0.38760
[249]	train-mlogloss:0.44087	eval-mlogloss:0.38711
[250]	train-mlogloss:0.44092	eval-mlogloss:0.38670
[251]	train-mlogloss:0.44103	eval-mlogloss:0.38646
[252]	train-mlogloss:0.44109	eval-mlogloss:0.38596
[253]	train-mlogloss:0.44101	eval-mlogloss:0.38611
[254]	train-mlogloss:0.44092	eval-mlogloss:0.38655
[255]	train-mlogloss:0.44088	eval-mlogloss:0.38669
[256]	train-mlogloss:0.44091	eval-mlogloss:0.38657
[257]	train-mlogloss:0.44087	eval-mlogloss:0.38633
[258]	train-mlogloss:0.44093	eval-mlogloss:0.38595
[259]	train-mlogloss:0.44105	eval-mlogloss:0.38566
[260]	train-mlogloss:0.44109	eval-mlogloss:0.38556
[261]	train-mlogloss:0.44102	eval-mlogloss:0.38577
[262]	train-mlogloss:0.44071	eval-mlogloss:0.38564
[263]	train-mlogloss:0.44063	eval-mlogloss:0.38579
[264]	train-mlogloss:0.44067	eval-mlogloss:0.38565
[265]	train-mlogloss:0.44061	eval-mlogloss:0.38587
[266]	train-mlogloss:0.44059	eval-mlogloss:0.38599
[267]	train-mlogloss:0.44064	eval-mlogloss:0.38577
[268]	train-mlogloss:0.44064	eval-mlogloss:0.38575
[269]	train-mlogloss:0.44040	eval-mlogloss:0.38581
[270]	train-mlogloss:0.44048	eval-mlogloss:0.38557
[271]	train-mlogloss:0.44042	eval-mlogloss:0.38578
[272]	train-mlogloss:0.44040	eval-mlogloss:0.38591
[273]	train-mlogloss:0.44041	eval-mlogloss:0.38618
[274]	train-mlogloss:0.44040	eval-mlogloss:0.38668
[275]	train-mlogloss:0.44043	eval-mlogloss:0.38690
[276]	train-mlogloss:0.44040	eval-mlogloss:0.38666
[277]	train-mlogloss:0.44037	eval-mlogloss:0.38663
[278]	train-mlogloss:0.43860	eval-mlogloss:0.38294
[279]	train-mlogloss:0.43865	eval-mlogloss:0.38308

```

[280] train-mlogloss:0.43858 eval-mlogloss:0.38226
[281] train-mlogloss:0.43860 eval-mlogloss:0.38216
[282] train-mlogloss:0.43833 eval-mlogloss:0.38136
[283] train-mlogloss:0.43822 eval-mlogloss:0.38152
[284] train-mlogloss:0.43816 eval-mlogloss:0.38186
[285] train-mlogloss:0.43814 eval-mlogloss:0.38234
[286] train-mlogloss:0.43592 eval-mlogloss:0.38014
[287] train-mlogloss:0.43588 eval-mlogloss:0.37936
[288] train-mlogloss:0.43587 eval-mlogloss:0.37968
[289] train-mlogloss:0.43564 eval-mlogloss:0.37914
[290] train-mlogloss:0.43529 eval-mlogloss:0.37910
[291] train-mlogloss:0.43534 eval-mlogloss:0.37923
[292] train-mlogloss:0.43534 eval-mlogloss:0.37928
[293] train-mlogloss:0.43536 eval-mlogloss:0.37926
[294] train-mlogloss:0.43541 eval-mlogloss:0.37934
[295] train-mlogloss:0.43535 eval-mlogloss:0.37889
[296] train-mlogloss:0.43536 eval-mlogloss:0.37926
[297] train-mlogloss:0.43537 eval-mlogloss:0.37949
[298] train-mlogloss:0.43542 eval-mlogloss:0.37967
[299] train-mlogloss:0.43529 eval-mlogloss:0.37898

```

[I 2025-12-06 14:55:15,063] Trial 10 finished with value: 0.9666666666666667 and parameters: {'lambda': 0.15855062547057477, 'alpha': 7.227778840637829e-06, 'eta': 0.08780784216645703, 'gamma': 2.0732539291037026e-08, 'max_depth': 7, 'min_child_weight': 10, 'subsample': 0.4454096415131273, 'colsample_bytree': 0.8360352992527816}. Best is trial 0 with value: 1.0.

```

[0] train-mlogloss:0.96174 eval-mlogloss:0.96400

```

[I 2025-12-06 14:55:15,115] Trial 11 pruned. Trial was pruned at iteration 1.

```

[0] train-mlogloss:1.07588 eval-mlogloss:1.07614
[1] train-mlogloss:1.05390 eval-mlogloss:1.05259
[2] train-mlogloss:1.03213 eval-mlogloss:1.02993
[3] train-mlogloss:1.01086 eval-mlogloss:1.00769
[4] train-mlogloss:0.99071 eval-mlogloss:0.98708
[5] train-mlogloss:0.97071 eval-mlogloss:0.96549
[6] train-mlogloss:0.95190 eval-mlogloss:0.94543
[7] train-mlogloss:0.93335 eval-mlogloss:0.92574
[8] train-mlogloss:0.91568 eval-mlogloss:0.90642
[9] train-mlogloss:0.89824 eval-mlogloss:0.88828
[10] train-mlogloss:0.88112 eval-mlogloss:0.87087
[11] train-mlogloss:0.86459 eval-mlogloss:0.85297
[12] train-mlogloss:0.84860 eval-mlogloss:0.83658
[13] train-mlogloss:0.83267 eval-mlogloss:0.81981
[14] train-mlogloss:0.81749 eval-mlogloss:0.80322
[15] train-mlogloss:0.80250 eval-mlogloss:0.78763
[16] train-mlogloss:0.78773 eval-mlogloss:0.77173
[17] train-mlogloss:0.77337 eval-mlogloss:0.75674
[18] train-mlogloss:0.75964 eval-mlogloss:0.74230

```


[19]	train-mlogloss:0.74609	eval-mlogloss:0.72807
[20]	train-mlogloss:0.73321	eval-mlogloss:0.71423
[21]	train-mlogloss:0.72030	eval-mlogloss:0.70026
[22]	train-mlogloss:0.70773	eval-mlogloss:0.68666
[23]	train-mlogloss:0.69540	eval-mlogloss:0.67381
[24]	train-mlogloss:0.68323	eval-mlogloss:0.66101
[25]	train-mlogloss:0.67150	eval-mlogloss:0.64825
[26]	train-mlogloss:0.66013	eval-mlogloss:0.63581
[27]	train-mlogloss:0.64888	eval-mlogloss:0.62386
[28]	train-mlogloss:0.63817	eval-mlogloss:0.61234
[29]	train-mlogloss:0.62770	eval-mlogloss:0.60077
[30]	train-mlogloss:0.61724	eval-mlogloss:0.58980
[31]	train-mlogloss:0.60699	eval-mlogloss:0.57894
[32]	train-mlogloss:0.59706	eval-mlogloss:0.56870
[33]	train-mlogloss:0.58710	eval-mlogloss:0.55786
[34]	train-mlogloss:0.57779	eval-mlogloss:0.54798
[35]	train-mlogloss:0.56878	eval-mlogloss:0.53799
[36]	train-mlogloss:0.55976	eval-mlogloss:0.52814
[37]	train-mlogloss:0.55077	eval-mlogloss:0.51856
[38]	train-mlogloss:0.54226	eval-mlogloss:0.50921
[39]	train-mlogloss:0.53374	eval-mlogloss:0.50037
[40]	train-mlogloss:0.52554	eval-mlogloss:0.49191
[41]	train-mlogloss:0.51734	eval-mlogloss:0.48293
[42]	train-mlogloss:0.50951	eval-mlogloss:0.47423
[43]	train-mlogloss:0.50158	eval-mlogloss:0.46583
[44]	train-mlogloss:0.49374	eval-mlogloss:0.45752
[45]	train-mlogloss:0.48618	eval-mlogloss:0.44933
[46]	train-mlogloss:0.47867	eval-mlogloss:0.44149
[47]	train-mlogloss:0.47160	eval-mlogloss:0.43399
[48]	train-mlogloss:0.46466	eval-mlogloss:0.42687
[49]	train-mlogloss:0.45770	eval-mlogloss:0.41918
[50]	train-mlogloss:0.45084	eval-mlogloss:0.41176
[51]	train-mlogloss:0.44432	eval-mlogloss:0.40495
[52]	train-mlogloss:0.43787	eval-mlogloss:0.39780
[53]	train-mlogloss:0.43178	eval-mlogloss:0.39095
[54]	train-mlogloss:0.42553	eval-mlogloss:0.38428
[55]	train-mlogloss:0.41957	eval-mlogloss:0.37804
[56]	train-mlogloss:0.41373	eval-mlogloss:0.37174
[57]	train-mlogloss:0.40794	eval-mlogloss:0.36575
[58]	train-mlogloss:0.40220	eval-mlogloss:0.35975
[59]	train-mlogloss:0.39655	eval-mlogloss:0.35373
[60]	train-mlogloss:0.39136	eval-mlogloss:0.34805
[61]	train-mlogloss:0.38609	eval-mlogloss:0.34217
[62]	train-mlogloss:0.38076	eval-mlogloss:0.33664
[63]	train-mlogloss:0.37575	eval-mlogloss:0.33108

[I 2025-12-06 14:55:15,407] Trial 12 pruned. Trial was pruned at iteration 64.

[0]	train-mlogloss:0.86724	eval-mlogloss:0.85759
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[I 2025-12-06 14:55:15,449] Trial 13 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.96930  eval-mlogloss:0.96549
[I 2025-12-06 14:55:15,492] Trial 14 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.85847  eval-mlogloss:0.85356
[I 2025-12-06 14:55:15,532] Trial 15 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.93506  eval-mlogloss:0.93113
[I 2025-12-06 14:55:15,576] Trial 16 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.83633  eval-mlogloss:0.82154
[I 2025-12-06 14:55:15,621] Trial 17 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.98985  eval-mlogloss:0.98631
[1]      train-mlogloss:0.94549  eval-mlogloss:0.93615
[2]      train-mlogloss:0.87224  eval-mlogloss:0.85896
[3]      train-mlogloss:0.78997  eval-mlogloss:0.77114
[I 2025-12-06 14:55:15,682] Trial 18 pruned. Trial was pruned at iteration 4.
[0]      train-mlogloss:0.85461  eval-mlogloss:0.84700
[I 2025-12-06 14:55:15,729] Trial 19 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.84452  eval-mlogloss:0.83066
[I 2025-12-06 14:55:15,773] Trial 20 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:1.07921  eval-mlogloss:1.07885
[1]      train-mlogloss:1.06431  eval-mlogloss:1.06128
[2]      train-mlogloss:1.04947  eval-mlogloss:1.04634
[3]      train-mlogloss:1.02513  eval-mlogloss:1.02061
[4]      train-mlogloss:1.00798  eval-mlogloss:1.00248
[5]      train-mlogloss:0.98686  eval-mlogloss:0.98034
[6]      train-mlogloss:0.97112  eval-mlogloss:0.96432
[7]      train-mlogloss:0.95568  eval-mlogloss:0.94865
[8]      train-mlogloss:0.94016  eval-mlogloss:0.93235
[9]      train-mlogloss:0.92614  eval-mlogloss:0.91741
[10]     train-mlogloss:0.91048  eval-mlogloss:0.90062
[11]     train-mlogloss:0.89168  eval-mlogloss:0.87997
[12]     train-mlogloss:0.87672  eval-mlogloss:0.86414
[13]     train-mlogloss:0.85927  eval-mlogloss:0.84529
[14]     train-mlogloss:0.84762  eval-mlogloss:0.83259
[15]     train-mlogloss:0.83342  eval-mlogloss:0.81782
[16]     train-mlogloss:0.82043  eval-mlogloss:0.80443
[17]     train-mlogloss:0.80966  eval-mlogloss:0.79311
[18]     train-mlogloss:0.79802  eval-mlogloss:0.78072
[19]     train-mlogloss:0.78289  eval-mlogloss:0.76428
[20]     train-mlogloss:0.77174  eval-mlogloss:0.75237
[21]     train-mlogloss:0.76093  eval-mlogloss:0.74091

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[22]	train-mlogloss:0.74778	eval-mlogloss:0.72634
[23]	train-mlogloss:0.73663	eval-mlogloss:0.71441
[24]	train-mlogloss:0.72305	eval-mlogloss:0.69914
[25]	train-mlogloss:0.71313	eval-mlogloss:0.68799
[26]	train-mlogloss:0.70015	eval-mlogloss:0.67343
[27]	train-mlogloss:0.69053	eval-mlogloss:0.66298
[28]	train-mlogloss:0.68096	eval-mlogloss:0.65335
[29]	train-mlogloss:0.67154	eval-mlogloss:0.64303
[30]	train-mlogloss:0.66049	eval-mlogloss:0.63180
[31]	train-mlogloss:0.65065	eval-mlogloss:0.62239
[32]	train-mlogloss:0.64289	eval-mlogloss:0.61529
[33]	train-mlogloss:0.63399	eval-mlogloss:0.60532
[34]	train-mlogloss:0.62492	eval-mlogloss:0.59599
[35]	train-mlogloss:0.61715	eval-mlogloss:0.58744
[36]	train-mlogloss:0.60939	eval-mlogloss:0.57947
[37]	train-mlogloss:0.60186	eval-mlogloss:0.57143
[38]	train-mlogloss:0.59443	eval-mlogloss:0.56325
[39]	train-mlogloss:0.58643	eval-mlogloss:0.55466
[40]	train-mlogloss:0.57701	eval-mlogloss:0.54537
[41]	train-mlogloss:0.56855	eval-mlogloss:0.53570
[42]	train-mlogloss:0.56215	eval-mlogloss:0.52850
[43]	train-mlogloss:0.55541	eval-mlogloss:0.52145
[44]	train-mlogloss:0.54826	eval-mlogloss:0.51358
[45]	train-mlogloss:0.54301	eval-mlogloss:0.50830
[46]	train-mlogloss:0.53642	eval-mlogloss:0.50147
[47]	train-mlogloss:0.53160	eval-mlogloss:0.49645
[48]	train-mlogloss:0.52500	eval-mlogloss:0.48871
[49]	train-mlogloss:0.51614	eval-mlogloss:0.47862
[50]	train-mlogloss:0.51031	eval-mlogloss:0.47230
[51]	train-mlogloss:0.50433	eval-mlogloss:0.46523
[52]	train-mlogloss:0.50037	eval-mlogloss:0.46115
[53]	train-mlogloss:0.49484	eval-mlogloss:0.45535
[54]	train-mlogloss:0.48941	eval-mlogloss:0.44928
[55]	train-mlogloss:0.48382	eval-mlogloss:0.44328
[56]	train-mlogloss:0.47792	eval-mlogloss:0.43696
[57]	train-mlogloss:0.47250	eval-mlogloss:0.43112
[58]	train-mlogloss:0.46775	eval-mlogloss:0.42631
[59]	train-mlogloss:0.46485	eval-mlogloss:0.42266
[60]	train-mlogloss:0.46296	eval-mlogloss:0.42087
[61]	train-mlogloss:0.46077	eval-mlogloss:0.41881
[62]	train-mlogloss:0.45557	eval-mlogloss:0.41353
[63]	train-mlogloss:0.45513	eval-mlogloss:0.41302

[I 2025-12-06 14:55:16,050] Trial 21 pruned. Trial was pruned at iteration 64.

[0]	train-mlogloss:1.02879	eval-mlogloss:1.03011
[1]	train-mlogloss:0.98186	eval-mlogloss:0.97461
[2]	train-mlogloss:0.92237	eval-mlogloss:0.91033
[3]	train-mlogloss:0.86319	eval-mlogloss:0.84864

[I 2025-12-06 14:55:16,106] Trial 22 pruned. Trial was pruned at iteration 4.

[0]	train-mlogloss:1.05012	eval-mlogloss:1.04908
[1]	train-mlogloss:1.02421	eval-mlogloss:1.01665
[2]	train-mlogloss:1.00064	eval-mlogloss:0.99226
[3]	train-mlogloss:0.94810	eval-mlogloss:0.93569

[I 2025-12-06 14:55:16,161] Trial 23 pruned. Trial was pruned at iteration 4.

[0]	train-mlogloss:0.91630	eval-mlogloss:0.91268
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[I 2025-12-06 14:55:16,206] Trial 24 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:0.98296	eval-mlogloss:0.97709
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[I 2025-12-06 14:55:16,248] Trial 25 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:1.02569	eval-mlogloss:1.02720
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[I 2025-12-06 14:55:16,291] Trial 26 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:0.96675	eval-mlogloss:0.96248
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[I 2025-12-06 14:55:16,333] Trial 27 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:0.88890	eval-mlogloss:0.89130
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[I 2025-12-06 14:55:16,374] Trial 28 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:0.81563	eval-mlogloss:0.80223
-----	------------------------	-----------------------

[I 2025-12-06 14:55:16,417] Trial 29 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:0.90458	eval-mlogloss:0.89395
-----	------------------------	-----------------------

[I 2025-12-06 14:55:16,461] Trial 30 pruned. Trial was pruned at iteration 1.

[0]	train-mlogloss:1.05058	eval-mlogloss:1.04928
[1]	train-mlogloss:1.01571	eval-mlogloss:1.01032
[2]	train-mlogloss:0.97608	eval-mlogloss:0.96879
[3]	train-mlogloss:0.94110	eval-mlogloss:0.93245

[I 2025-12-06 14:55:16,512] Trial 31 pruned. Trial was pruned at iteration 4.

[0]	train-mlogloss:1.04213	eval-mlogloss:1.04048
[1]	train-mlogloss:1.00175	eval-mlogloss:0.99536
[2]	train-mlogloss:0.95648	eval-mlogloss:0.94792
[3]	train-mlogloss:0.91470	eval-mlogloss:0.90320

[I 2025-12-06 14:55:16,566] Trial 32 pruned. Trial was pruned at iteration 4.

[0]	train-mlogloss:1.05472	eval-mlogloss:1.05591
[1]	train-mlogloss:1.02120	eval-mlogloss:1.01719
[2]	train-mlogloss:0.98151	eval-mlogloss:0.97406
[3]	train-mlogloss:0.94092	eval-mlogloss:0.93175

[I 2025-12-06 14:55:16,621] Trial 33 pruned. Trial was pruned at iteration 4.

[0]	train-mlogloss:1.01699	eval-mlogloss:1.01551
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[I 2025-12-06 14:55:16,665] Trial 34 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.79975  eval-mlogloss:0.80372

[I 2025-12-06 14:55:16,710] Trial 35 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:1.05304  eval-mlogloss:1.05264
[1]      train-mlogloss:1.01072  eval-mlogloss:1.00769
[2]      train-mlogloss:0.96947  eval-mlogloss:0.96525
[3]      train-mlogloss:0.93112  eval-mlogloss:0.92609

[I 2025-12-06 14:55:16,768] Trial 36 pruned. Trial was pruned at iteration 4.
[0]      train-mlogloss:1.00577  eval-mlogloss:1.00225

[I 2025-12-06 14:55:16,817] Trial 37 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.97299  eval-mlogloss:0.96784

[I 2025-12-06 14:55:16,865] Trial 38 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.81179  eval-mlogloss:0.81482

[I 2025-12-06 14:55:16,914] Trial 39 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:0.88958  eval-mlogloss:0.88513

[I 2025-12-06 14:55:16,961] Trial 40 pruned. Trial was pruned at iteration 1.
[0]      train-mlogloss:1.08649  eval-mlogloss:1.08653
[1]      train-mlogloss:1.07706  eval-mlogloss:1.07543
[2]      train-mlogloss:1.06764  eval-mlogloss:1.06593
[3]      train-mlogloss:1.05210  eval-mlogloss:1.04951
[4]      train-mlogloss:1.04097  eval-mlogloss:1.03774
[5]      train-mlogloss:1.02714  eval-mlogloss:1.02324
[6]      train-mlogloss:1.01669  eval-mlogloss:1.01264
[7]      train-mlogloss:1.00672  eval-mlogloss:1.00261
[8]      train-mlogloss:0.99623  eval-mlogloss:0.99159
[9]      train-mlogloss:0.98663  eval-mlogloss:0.98133
[10]     train-mlogloss:0.97372  eval-mlogloss:0.96788
[11]     train-mlogloss:0.96040  eval-mlogloss:0.95357
[12]     train-mlogloss:0.95009  eval-mlogloss:0.94342
[13]     train-mlogloss:0.93787  eval-mlogloss:0.93024
[14]     train-mlogloss:0.92893  eval-mlogloss:0.92121
[15]     train-mlogloss:0.91879  eval-mlogloss:0.91065
[16]     train-mlogloss:0.90757  eval-mlogloss:0.89917
[17]     train-mlogloss:0.89872  eval-mlogloss:0.88901
[18]     train-mlogloss:0.88708  eval-mlogloss:0.87705
[19]     train-mlogloss:0.87515  eval-mlogloss:0.86450
[20]     train-mlogloss:0.86689  eval-mlogloss:0.85565
[21]     train-mlogloss:0.85882  eval-mlogloss:0.84708
[22]     train-mlogloss:0.84858  eval-mlogloss:0.83590
[23]     train-mlogloss:0.83990  eval-mlogloss:0.82691
[24]     train-mlogloss:0.82967  eval-mlogloss:0.81545

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[25]	train-mlogloss:0.82200	eval-mlogloss:0.80723
[26]	train-mlogloss:0.81198	eval-mlogloss:0.79605
[27]	train-mlogloss:0.80445	eval-mlogloss:0.78789
[28]	train-mlogloss:0.79688	eval-mlogloss:0.77990
[29]	train-mlogloss:0.78938	eval-mlogloss:0.77171
[30]	train-mlogloss:0.78073	eval-mlogloss:0.76287
[31]	train-mlogloss:0.77290	eval-mlogloss:0.75531
[32]	train-mlogloss:0.76414	eval-mlogloss:0.74666
[33]	train-mlogloss:0.75692	eval-mlogloss:0.73836
[34]	train-mlogloss:0.74793	eval-mlogloss:0.72909
[35]	train-mlogloss:0.74157	eval-mlogloss:0.72225
[36]	train-mlogloss:0.73518	eval-mlogloss:0.71565
[37]	train-mlogloss:0.72710	eval-mlogloss:0.70697
[38]	train-mlogloss:0.72075	eval-mlogloss:0.70031
[39]	train-mlogloss:0.71414	eval-mlogloss:0.69321
[40]	train-mlogloss:0.70603	eval-mlogloss:0.68532
[41]	train-mlogloss:0.69765	eval-mlogloss:0.67623
[42]	train-mlogloss:0.69203	eval-mlogloss:0.66989
[43]	train-mlogloss:0.68620	eval-mlogloss:0.66374
[44]	train-mlogloss:0.68008	eval-mlogloss:0.65701
[45]	train-mlogloss:0.67273	eval-mlogloss:0.64992
[46]	train-mlogloss:0.66658	eval-mlogloss:0.64354
[47]	train-mlogloss:0.66058	eval-mlogloss:0.63734
[48]	train-mlogloss:0.65478	eval-mlogloss:0.63094
[49]	train-mlogloss:0.64693	eval-mlogloss:0.62237
[50]	train-mlogloss:0.64136	eval-mlogloss:0.61618
[51]	train-mlogloss:0.63608	eval-mlogloss:0.60996
[52]	train-mlogloss:0.63131	eval-mlogloss:0.60525
[53]	train-mlogloss:0.62604	eval-mlogloss:0.59977
[54]	train-mlogloss:0.62064	eval-mlogloss:0.59406
[55]	train-mlogloss:0.61393	eval-mlogloss:0.58685
[56]	train-mlogloss:0.60754	eval-mlogloss:0.57953
[57]	train-mlogloss:0.60264	eval-mlogloss:0.57424
[58]	train-mlogloss:0.59797	eval-mlogloss:0.56935
[59]	train-mlogloss:0.59303	eval-mlogloss:0.56378
[60]	train-mlogloss:0.58910	eval-mlogloss:0.55964
[61]	train-mlogloss:0.58649	eval-mlogloss:0.55691
[62]	train-mlogloss:0.58145	eval-mlogloss:0.55182
[63]	train-mlogloss:0.57762	eval-mlogloss:0.54820
[64]	train-mlogloss:0.57381	eval-mlogloss:0.54399
[65]	train-mlogloss:0.56924	eval-mlogloss:0.53904
[66]	train-mlogloss:0.56497	eval-mlogloss:0.53442
[67]	train-mlogloss:0.56130	eval-mlogloss:0.53042
[68]	train-mlogloss:0.55718	eval-mlogloss:0.52587
[69]	train-mlogloss:0.55351	eval-mlogloss:0.52159
[70]	train-mlogloss:0.54956	eval-mlogloss:0.51734
[71]	train-mlogloss:0.54579	eval-mlogloss:0.51362
[72]	train-mlogloss:0.54237	eval-mlogloss:0.51003

[73]	train-mlogloss:0.53703	eval-mlogloss:0.50426
[74]	train-mlogloss:0.53286	eval-mlogloss:0.49986
[75]	train-mlogloss:0.52808	eval-mlogloss:0.49439
[76]	train-mlogloss:0.52303	eval-mlogloss:0.48889
[77]	train-mlogloss:0.51831	eval-mlogloss:0.48349
[78]	train-mlogloss:0.51434	eval-mlogloss:0.47937
[79]	train-mlogloss:0.51145	eval-mlogloss:0.47600
[80]	train-mlogloss:0.50894	eval-mlogloss:0.47327
[81]	train-mlogloss:0.50538	eval-mlogloss:0.46948
[82]	train-mlogloss:0.50174	eval-mlogloss:0.46558
[83]	train-mlogloss:0.49809	eval-mlogloss:0.46146
[84]	train-mlogloss:0.49469	eval-mlogloss:0.45793
[85]	train-mlogloss:0.49159	eval-mlogloss:0.45510
[86]	train-mlogloss:0.48834	eval-mlogloss:0.45118
[87]	train-mlogloss:0.48512	eval-mlogloss:0.44782
[88]	train-mlogloss:0.48199	eval-mlogloss:0.44461
[89]	train-mlogloss:0.47865	eval-mlogloss:0.44118
[90]	train-mlogloss:0.47533	eval-mlogloss:0.43734
[91]	train-mlogloss:0.47254	eval-mlogloss:0.43456
[92]	train-mlogloss:0.46900	eval-mlogloss:0.43092
[93]	train-mlogloss:0.46724	eval-mlogloss:0.42918
[94]	train-mlogloss:0.46593	eval-mlogloss:0.42755
[95]	train-mlogloss:0.46485	eval-mlogloss:0.42657
[96]	train-mlogloss:0.46317	eval-mlogloss:0.42446
[97]	train-mlogloss:0.46017	eval-mlogloss:0.42105
[98]	train-mlogloss:0.45720	eval-mlogloss:0.41779
[99]	train-mlogloss:0.45449	eval-mlogloss:0.41482
[100]	train-mlogloss:0.45168	eval-mlogloss:0.41217
[101]	train-mlogloss:0.44856	eval-mlogloss:0.40901
[102]	train-mlogloss:0.44613	eval-mlogloss:0.40600
[103]	train-mlogloss:0.44350	eval-mlogloss:0.40326
[104]	train-mlogloss:0.44107	eval-mlogloss:0.40072
[105]	train-mlogloss:0.43991	eval-mlogloss:0.39939
[106]	train-mlogloss:0.43827	eval-mlogloss:0.39732
[107]	train-mlogloss:0.43615	eval-mlogloss:0.39509
[108]	train-mlogloss:0.43483	eval-mlogloss:0.39381
[109]	train-mlogloss:0.43223	eval-mlogloss:0.39062
[110]	train-mlogloss:0.42941	eval-mlogloss:0.38728
[111]	train-mlogloss:0.42673	eval-mlogloss:0.38426
[112]	train-mlogloss:0.42563	eval-mlogloss:0.38282
[113]	train-mlogloss:0.42434	eval-mlogloss:0.38102
[114]	train-mlogloss:0.42367	eval-mlogloss:0.38021
[115]	train-mlogloss:0.42204	eval-mlogloss:0.37820
[116]	train-mlogloss:0.42184	eval-mlogloss:0.37798
[117]	train-mlogloss:0.41913	eval-mlogloss:0.37524
[118]	train-mlogloss:0.41898	eval-mlogloss:0.37515
[119]	train-mlogloss:0.41628	eval-mlogloss:0.37229
[120]	train-mlogloss:0.41500	eval-mlogloss:0.37097

[121]	train-mlogloss:0.41314	eval-mlogloss:0.36897
[122]	train-mlogloss:0.41088	eval-mlogloss:0.36638
[123]	train-mlogloss:0.41018	eval-mlogloss:0.36562
[124]	train-mlogloss:0.40889	eval-mlogloss:0.36400
[125]	train-mlogloss:0.40740	eval-mlogloss:0.36214
[126]	train-mlogloss:0.40643	eval-mlogloss:0.36093
[127]	train-mlogloss:0.40615	eval-mlogloss:0.36060
[128]	train-mlogloss:0.40593	eval-mlogloss:0.36039
[129]	train-mlogloss:0.40475	eval-mlogloss:0.35908
[130]	train-mlogloss:0.40367	eval-mlogloss:0.35806
[131]	train-mlogloss:0.40367	eval-mlogloss:0.35808
[132]	train-mlogloss:0.40170	eval-mlogloss:0.35590
[133]	train-mlogloss:0.40051	eval-mlogloss:0.35449
[134]	train-mlogloss:0.40048	eval-mlogloss:0.35447
[135]	train-mlogloss:0.39809	eval-mlogloss:0.35201
[136]	train-mlogloss:0.39713	eval-mlogloss:0.35099
[137]	train-mlogloss:0.39509	eval-mlogloss:0.34881
[138]	train-mlogloss:0.39392	eval-mlogloss:0.34764
[139]	train-mlogloss:0.39356	eval-mlogloss:0.34717
[140]	train-mlogloss:0.39345	eval-mlogloss:0.34704
[141]	train-mlogloss:0.39209	eval-mlogloss:0.34550
[142]	train-mlogloss:0.39003	eval-mlogloss:0.34330
[143]	train-mlogloss:0.38909	eval-mlogloss:0.34223
[144]	train-mlogloss:0.38891	eval-mlogloss:0.34179
[145]	train-mlogloss:0.38881	eval-mlogloss:0.34173
[146]	train-mlogloss:0.38861	eval-mlogloss:0.34158
[147]	train-mlogloss:0.38751	eval-mlogloss:0.33998
[148]	train-mlogloss:0.38627	eval-mlogloss:0.33860
[149]	train-mlogloss:0.38624	eval-mlogloss:0.33864
[150]	train-mlogloss:0.38609	eval-mlogloss:0.33827
[151]	train-mlogloss:0.38609	eval-mlogloss:0.33827
[152]	train-mlogloss:0.38608	eval-mlogloss:0.33825
[153]	train-mlogloss:0.38406	eval-mlogloss:0.33621
[154]	train-mlogloss:0.38387	eval-mlogloss:0.33603
[155]	train-mlogloss:0.38382	eval-mlogloss:0.33608
[156]	train-mlogloss:0.38286	eval-mlogloss:0.33483
[157]	train-mlogloss:0.38286	eval-mlogloss:0.33481
[158]	train-mlogloss:0.38197	eval-mlogloss:0.33364
[159]	train-mlogloss:0.38185	eval-mlogloss:0.33355
[160]	train-mlogloss:0.38174	eval-mlogloss:0.33326
[161]	train-mlogloss:0.38161	eval-mlogloss:0.33316
[162]	train-mlogloss:0.38160	eval-mlogloss:0.33315
[163]	train-mlogloss:0.38157	eval-mlogloss:0.33313
[164]	train-mlogloss:0.38139	eval-mlogloss:0.33273
[165]	train-mlogloss:0.38039	eval-mlogloss:0.33126
[166]	train-mlogloss:0.37933	eval-mlogloss:0.32988
[167]	train-mlogloss:0.37929	eval-mlogloss:0.32993
[168]	train-mlogloss:0.37857	eval-mlogloss:0.32911

[169]	train-mlogloss:0.37839	eval-mlogloss:0.32883
[170]	train-mlogloss:0.37834	eval-mlogloss:0.32881
[171]	train-mlogloss:0.37786	eval-mlogloss:0.32820
[172]	train-mlogloss:0.37779	eval-mlogloss:0.32802
[173]	train-mlogloss:0.37697	eval-mlogloss:0.32708
[174]	train-mlogloss:0.37602	eval-mlogloss:0.32590
[175]	train-mlogloss:0.37507	eval-mlogloss:0.32474
[176]	train-mlogloss:0.37506	eval-mlogloss:0.32473
[177]	train-mlogloss:0.37321	eval-mlogloss:0.32259
[178]	train-mlogloss:0.37311	eval-mlogloss:0.32250
[179]	train-mlogloss:0.37299	eval-mlogloss:0.32219
[180]	train-mlogloss:0.37286	eval-mlogloss:0.32187
[181]	train-mlogloss:0.37279	eval-mlogloss:0.32181
[182]	train-mlogloss:0.37280	eval-mlogloss:0.32181
[183]	train-mlogloss:0.37267	eval-mlogloss:0.32154
[184]	train-mlogloss:0.37140	eval-mlogloss:0.32031
[185]	train-mlogloss:0.37128	eval-mlogloss:0.32015
[186]	train-mlogloss:0.37100	eval-mlogloss:0.31977
[187]	train-mlogloss:0.36997	eval-mlogloss:0.31867
[188]	train-mlogloss:0.36986	eval-mlogloss:0.31844
[189]	train-mlogloss:0.36982	eval-mlogloss:0.31846
[190]	train-mlogloss:0.36972	eval-mlogloss:0.31825
[191]	train-mlogloss:0.36956	eval-mlogloss:0.31784
[192]	train-mlogloss:0.36952	eval-mlogloss:0.31782
[193]	train-mlogloss:0.36906	eval-mlogloss:0.31702
[194]	train-mlogloss:0.36806	eval-mlogloss:0.31587
[195]	train-mlogloss:0.36807	eval-mlogloss:0.31586
[196]	train-mlogloss:0.36802	eval-mlogloss:0.31586
[197]	train-mlogloss:0.36801	eval-mlogloss:0.31585
[198]	train-mlogloss:0.36797	eval-mlogloss:0.31588
[199]	train-mlogloss:0.36716	eval-mlogloss:0.31481
[200]	train-mlogloss:0.36709	eval-mlogloss:0.31475
[201]	train-mlogloss:0.36525	eval-mlogloss:0.31291
[202]	train-mlogloss:0.36520	eval-mlogloss:0.31288
[203]	train-mlogloss:0.36520	eval-mlogloss:0.31290
[204]	train-mlogloss:0.36501	eval-mlogloss:0.31268
[205]	train-mlogloss:0.36491	eval-mlogloss:0.31239
[206]	train-mlogloss:0.36486	eval-mlogloss:0.31234
[207]	train-mlogloss:0.36487	eval-mlogloss:0.31238
[208]	train-mlogloss:0.36472	eval-mlogloss:0.31229
[209]	train-mlogloss:0.36365	eval-mlogloss:0.31081
[210]	train-mlogloss:0.36352	eval-mlogloss:0.31046
[211]	train-mlogloss:0.36325	eval-mlogloss:0.31009
[212]	train-mlogloss:0.36320	eval-mlogloss:0.31005
[213]	train-mlogloss:0.36313	eval-mlogloss:0.31001
[214]	train-mlogloss:0.36311	eval-mlogloss:0.31001
[215]	train-mlogloss:0.36228	eval-mlogloss:0.30904
[216]	train-mlogloss:0.36141	eval-mlogloss:0.30778

[217]	train-mlogloss:0.36133	eval-mlogloss:0.30755
[218]	train-mlogloss:0.36055	eval-mlogloss:0.30654
[219]	train-mlogloss:0.35940	eval-mlogloss:0.30551
[220]	train-mlogloss:0.35931	eval-mlogloss:0.30550
[221]	train-mlogloss:0.35902	eval-mlogloss:0.30521
[222]	train-mlogloss:0.35897	eval-mlogloss:0.30520
[223]	train-mlogloss:0.35892	eval-mlogloss:0.30520
[224]	train-mlogloss:0.35857	eval-mlogloss:0.30469
[225]	train-mlogloss:0.35766	eval-mlogloss:0.30367
[226]	train-mlogloss:0.35768	eval-mlogloss:0.30363
[227]	train-mlogloss:0.35761	eval-mlogloss:0.30359
[228]	train-mlogloss:0.35757	eval-mlogloss:0.30359
[229]	train-mlogloss:0.35681	eval-mlogloss:0.30281
[230]	train-mlogloss:0.35670	eval-mlogloss:0.30249
[231]	train-mlogloss:0.35662	eval-mlogloss:0.30229
[232]	train-mlogloss:0.35658	eval-mlogloss:0.30232
[233]	train-mlogloss:0.35655	eval-mlogloss:0.30231
[234]	train-mlogloss:0.35654	eval-mlogloss:0.30235
[235]	train-mlogloss:0.35653	eval-mlogloss:0.30237
[236]	train-mlogloss:0.35648	eval-mlogloss:0.30232
[237]	train-mlogloss:0.35647	eval-mlogloss:0.30232
[238]	train-mlogloss:0.35638	eval-mlogloss:0.30233
[239]	train-mlogloss:0.35632	eval-mlogloss:0.30215
[240]	train-mlogloss:0.35538	eval-mlogloss:0.30094
[241]	train-mlogloss:0.35461	eval-mlogloss:0.29992
[242]	train-mlogloss:0.35461	eval-mlogloss:0.29995
[243]	train-mlogloss:0.35456	eval-mlogloss:0.29993
[244]	train-mlogloss:0.35456	eval-mlogloss:0.29994
[245]	train-mlogloss:0.35448	eval-mlogloss:0.29985
[246]	train-mlogloss:0.35449	eval-mlogloss:0.29978
[247]	train-mlogloss:0.35446	eval-mlogloss:0.29976
[248]	train-mlogloss:0.35446	eval-mlogloss:0.29977
[249]	train-mlogloss:0.35446	eval-mlogloss:0.29973
[250]	train-mlogloss:0.35440	eval-mlogloss:0.29953
[251]	train-mlogloss:0.35440	eval-mlogloss:0.29950
[252]	train-mlogloss:0.35431	eval-mlogloss:0.29940
[253]	train-mlogloss:0.35425	eval-mlogloss:0.29932
[254]	train-mlogloss:0.35417	eval-mlogloss:0.29927
[255]	train-mlogloss:0.35411	eval-mlogloss:0.29922

[I 2025-12-06 14:55:17,930] Trial 41 pruned. Trial was pruned at iteration 256.

[0]	train-mlogloss:1.08865	eval-mlogloss:1.08880
[1]	train-mlogloss:1.08093	eval-mlogloss:1.07972
[2]	train-mlogloss:1.07292	eval-mlogloss:1.07135
[3]	train-mlogloss:1.06004	eval-mlogloss:1.05774
[4]	train-mlogloss:1.05075	eval-mlogloss:1.04792
[5]	train-mlogloss:1.03922	eval-mlogloss:1.03583
[6]	train-mlogloss:1.03046	eval-mlogloss:1.02695

[7]	train-mlogloss:1.02208	eval-mlogloss:1.01852
[8]	train-mlogloss:1.01324	eval-mlogloss:1.00923
[9]	train-mlogloss:1.00513	eval-mlogloss:1.00056
[10]	train-mlogloss:0.99448	eval-mlogloss:0.98940
[11]	train-mlogloss:0.98319	eval-mlogloss:0.97728
[12]	train-mlogloss:0.97439	eval-mlogloss:0.96861
[13]	train-mlogloss:0.96345	eval-mlogloss:0.95713
[14]	train-mlogloss:0.95579	eval-mlogloss:0.94938
[15]	train-mlogloss:0.94708	eval-mlogloss:0.94031
[16]	train-mlogloss:0.93895	eval-mlogloss:0.93191
[17]	train-mlogloss:0.93227	eval-mlogloss:0.92486
[18]	train-mlogloss:0.92217	eval-mlogloss:0.91448
[19]	train-mlogloss:0.91181	eval-mlogloss:0.90359
[20]	train-mlogloss:0.90459	eval-mlogloss:0.89586
[21]	train-mlogloss:0.89751	eval-mlogloss:0.88834
[22]	train-mlogloss:0.88852	eval-mlogloss:0.87854
[23]	train-mlogloss:0.88089	eval-mlogloss:0.87064
[24]	train-mlogloss:0.87189	eval-mlogloss:0.86059
[25]	train-mlogloss:0.86510	eval-mlogloss:0.85331
[26]	train-mlogloss:0.85590	eval-mlogloss:0.84323
[27]	train-mlogloss:0.84915	eval-mlogloss:0.83624
[28]	train-mlogloss:0.84241	eval-mlogloss:0.82912
[29]	train-mlogloss:0.83569	eval-mlogloss:0.82179
[30]	train-mlogloss:0.82795	eval-mlogloss:0.81388
[31]	train-mlogloss:0.82042	eval-mlogloss:0.80645
[32]	train-mlogloss:0.81234	eval-mlogloss:0.79828
[33]	train-mlogloss:0.80373	eval-mlogloss:0.78864
[34]	train-mlogloss:0.79566	eval-mlogloss:0.78032
[35]	train-mlogloss:0.78993	eval-mlogloss:0.77414
[36]	train-mlogloss:0.78413	eval-mlogloss:0.76814
[37]	train-mlogloss:0.77684	eval-mlogloss:0.76032
[38]	train-mlogloss:0.77108	eval-mlogloss:0.75427
[39]	train-mlogloss:0.76509	eval-mlogloss:0.74786
[40]	train-mlogloss:0.75772	eval-mlogloss:0.74067
[41]	train-mlogloss:0.75009	eval-mlogloss:0.73242
[42]	train-mlogloss:0.74442	eval-mlogloss:0.72629
[43]	train-mlogloss:0.73905	eval-mlogloss:0.72063
[44]	train-mlogloss:0.73344	eval-mlogloss:0.71445
[45]	train-mlogloss:0.72652	eval-mlogloss:0.70752
[46]	train-mlogloss:0.72084	eval-mlogloss:0.70164
[47]	train-mlogloss:0.71524	eval-mlogloss:0.69586
[48]	train-mlogloss:0.70994	eval-mlogloss:0.69003
[49]	train-mlogloss:0.70275	eval-mlogloss:0.68219
[50]	train-mlogloss:0.69767	eval-mlogloss:0.67659
[51]	train-mlogloss:0.69270	eval-mlogloss:0.67055
[52]	train-mlogloss:0.68688	eval-mlogloss:0.66443
[53]	train-mlogloss:0.68181	eval-mlogloss:0.65894
[54]	train-mlogloss:0.67668	eval-mlogloss:0.65355

[55]	train-mlogloss:0.67030	eval-mlogloss:0.64678
[56]	train-mlogloss:0.66435	eval-mlogloss:0.64000
[57]	train-mlogloss:0.65975	eval-mlogloss:0.63504
[58]	train-mlogloss:0.65534	eval-mlogloss:0.63042
[59]	train-mlogloss:0.65068	eval-mlogloss:0.62518
[60]	train-mlogloss:0.64693	eval-mlogloss:0.62124
[61]	train-mlogloss:0.64346	eval-mlogloss:0.61757
[62]	train-mlogloss:0.63861	eval-mlogloss:0.61271
[63]	train-mlogloss:0.63495	eval-mlogloss:0.60909
[64]	train-mlogloss:0.63132	eval-mlogloss:0.60510
[65]	train-mlogloss:0.62689	eval-mlogloss:0.60034
[66]	train-mlogloss:0.62179	eval-mlogloss:0.59448
[67]	train-mlogloss:0.61820	eval-mlogloss:0.59059
[68]	train-mlogloss:0.61428	eval-mlogloss:0.58628
[69]	train-mlogloss:0.61027	eval-mlogloss:0.58189
[70]	train-mlogloss:0.60649	eval-mlogloss:0.57774
[71]	train-mlogloss:0.60287	eval-mlogloss:0.57416
[72]	train-mlogloss:0.59956	eval-mlogloss:0.57069
[73]	train-mlogloss:0.59429	eval-mlogloss:0.56467
[74]	train-mlogloss:0.59026	eval-mlogloss:0.56045
[75]	train-mlogloss:0.58568	eval-mlogloss:0.55521
[76]	train-mlogloss:0.58088	eval-mlogloss:0.55001
[77]	train-mlogloss:0.57629	eval-mlogloss:0.54497
[78]	train-mlogloss:0.57246	eval-mlogloss:0.54100
[79]	train-mlogloss:0.56926	eval-mlogloss:0.53736
[80]	train-mlogloss:0.56565	eval-mlogloss:0.53341
[81]	train-mlogloss:0.56207	eval-mlogloss:0.52965
[82]	train-mlogloss:0.55839	eval-mlogloss:0.52574
[83]	train-mlogloss:0.55467	eval-mlogloss:0.52160
[84]	train-mlogloss:0.55134	eval-mlogloss:0.51814
[85]	train-mlogloss:0.54795	eval-mlogloss:0.51479
[86]	train-mlogloss:0.54450	eval-mlogloss:0.51075
[87]	train-mlogloss:0.54132	eval-mlogloss:0.50733
[88]	train-mlogloss:0.53822	eval-mlogloss:0.50416
[89]	train-mlogloss:0.53477	eval-mlogloss:0.50051
[90]	train-mlogloss:0.53149	eval-mlogloss:0.49673
[91]	train-mlogloss:0.52864	eval-mlogloss:0.49419
[92]	train-mlogloss:0.52521	eval-mlogloss:0.49064
[93]	train-mlogloss:0.52348	eval-mlogloss:0.48893
[94]	train-mlogloss:0.52041	eval-mlogloss:0.48565
[95]	train-mlogloss:0.51782	eval-mlogloss:0.48300
[96]	train-mlogloss:0.51393	eval-mlogloss:0.47888
[97]	train-mlogloss:0.51041	eval-mlogloss:0.47473
[98]	train-mlogloss:0.50754	eval-mlogloss:0.47158
[99]	train-mlogloss:0.50471	eval-mlogloss:0.46847
[100]	train-mlogloss:0.50183	eval-mlogloss:0.46559
[101]	train-mlogloss:0.49878	eval-mlogloss:0.46251
[102]	train-mlogloss:0.49580	eval-mlogloss:0.45890

[103]	train-mlogloss:0.49313	eval-mlogloss:0.45591
[104]	train-mlogloss:0.48952	eval-mlogloss:0.45190
[105]	train-mlogloss:0.48729	eval-mlogloss:0.44952
[106]	train-mlogloss:0.48465	eval-mlogloss:0.44640
[107]	train-mlogloss:0.48216	eval-mlogloss:0.44368
[108]	train-mlogloss:0.47932	eval-mlogloss:0.44051
[109]	train-mlogloss:0.47642	eval-mlogloss:0.43724
[110]	train-mlogloss:0.47368	eval-mlogloss:0.43402
[111]	train-mlogloss:0.47084	eval-mlogloss:0.43087
[112]	train-mlogloss:0.46971	eval-mlogloss:0.42944
[113]	train-mlogloss:0.46842	eval-mlogloss:0.42767
[114]	train-mlogloss:0.46613	eval-mlogloss:0.42506
[115]	train-mlogloss:0.46334	eval-mlogloss:0.42175
[116]	train-mlogloss:0.46287	eval-mlogloss:0.42120
[117]	train-mlogloss:0.46013	eval-mlogloss:0.41829
[118]	train-mlogloss:0.45810	eval-mlogloss:0.41615
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[121]	train-mlogloss:0.45058	eval-mlogloss:0.40828
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[123]	train-mlogloss:0.44670	eval-mlogloss:0.40415
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[125]	train-mlogloss:0.44287	eval-mlogloss:0.39945
[126]	train-mlogloss:0.44184	eval-mlogloss:0.39831
[127]	train-mlogloss:0.44032	eval-mlogloss:0.39667
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[129]	train-mlogloss:0.43783	eval-mlogloss:0.39379
[130]	train-mlogloss:0.43661	eval-mlogloss:0.39261
[131]	train-mlogloss:0.43598	eval-mlogloss:0.39197
[132]	train-mlogloss:0.43386	eval-mlogloss:0.38977
[133]	train-mlogloss:0.43274	eval-mlogloss:0.38847
[134]	train-mlogloss:0.43269	eval-mlogloss:0.38839
[135]	train-mlogloss:0.43047	eval-mlogloss:0.38612
[136]	train-mlogloss:0.42939	eval-mlogloss:0.38463
[137]	train-mlogloss:0.42731	eval-mlogloss:0.38228
[138]	train-mlogloss:0.42607	eval-mlogloss:0.38087
[139]	train-mlogloss:0.42517	eval-mlogloss:0.37975
[140]	train-mlogloss:0.42508	eval-mlogloss:0.37965
[141]	train-mlogloss:0.42380	eval-mlogloss:0.37817
[142]	train-mlogloss:0.42176	eval-mlogloss:0.37606
[143]	train-mlogloss:0.42071	eval-mlogloss:0.37484
[144]	train-mlogloss:0.41871	eval-mlogloss:0.37259
[145]	train-mlogloss:0.41862	eval-mlogloss:0.37253
[146]	train-mlogloss:0.41835	eval-mlogloss:0.37219
[147]	train-mlogloss:0.41731	eval-mlogloss:0.37070
[148]	train-mlogloss:0.41497	eval-mlogloss:0.36812
[149]	train-mlogloss:0.41406	eval-mlogloss:0.36713
[150]	train-mlogloss:0.41277	eval-mlogloss:0.36567

[151]	train-mlogloss:0.41100	eval-mlogloss:0.36362
[152]	train-mlogloss:0.41097	eval-mlogloss:0.36361
[153]	train-mlogloss:0.40891	eval-mlogloss:0.36157
[154]	train-mlogloss:0.40790	eval-mlogloss:0.36058
[155]	train-mlogloss:0.40724	eval-mlogloss:0.35988
[156]	train-mlogloss:0.40529	eval-mlogloss:0.35781
[157]	train-mlogloss:0.40527	eval-mlogloss:0.35777
[158]	train-mlogloss:0.40439	eval-mlogloss:0.35683
[159]	train-mlogloss:0.40353	eval-mlogloss:0.35592
[160]	train-mlogloss:0.40302	eval-mlogloss:0.35511
[161]	train-mlogloss:0.40279	eval-mlogloss:0.35484
[162]	train-mlogloss:0.40275	eval-mlogloss:0.35480
[163]	train-mlogloss:0.40189	eval-mlogloss:0.35382
[164]	train-mlogloss:0.40170	eval-mlogloss:0.35343
[165]	train-mlogloss:0.40079	eval-mlogloss:0.35210
[166]	train-mlogloss:0.39971	eval-mlogloss:0.35073
[167]	train-mlogloss:0.39957	eval-mlogloss:0.35064
[168]	train-mlogloss:0.39892	eval-mlogloss:0.34991
[169]	train-mlogloss:0.39798	eval-mlogloss:0.34885
[170]	train-mlogloss:0.39718	eval-mlogloss:0.34795
[171]	train-mlogloss:0.39619	eval-mlogloss:0.34685
[172]	train-mlogloss:0.39594	eval-mlogloss:0.34641
[173]	train-mlogloss:0.39430	eval-mlogloss:0.34443
[174]	train-mlogloss:0.39332	eval-mlogloss:0.34324
[175]	train-mlogloss:0.39248	eval-mlogloss:0.34223
[176]	train-mlogloss:0.39245	eval-mlogloss:0.34220
[177]	train-mlogloss:0.39078	eval-mlogloss:0.34029
[178]	train-mlogloss:0.38987	eval-mlogloss:0.33918
[179]	train-mlogloss:0.38905	eval-mlogloss:0.33805
[180]	train-mlogloss:0.38891	eval-mlogloss:0.33775
[181]	train-mlogloss:0.38886	eval-mlogloss:0.33769
[182]	train-mlogloss:0.38829	eval-mlogloss:0.33711
[183]	train-mlogloss:0.38718	eval-mlogloss:0.33580
[184]	train-mlogloss:0.38606	eval-mlogloss:0.33469
[185]	train-mlogloss:0.38592	eval-mlogloss:0.33453
[186]	train-mlogloss:0.38569	eval-mlogloss:0.33422
[187]	train-mlogloss:0.38467	eval-mlogloss:0.33320
[188]	train-mlogloss:0.38456	eval-mlogloss:0.33298
[189]	train-mlogloss:0.38452	eval-mlogloss:0.33300
[190]	train-mlogloss:0.38432	eval-mlogloss:0.33280
[191]	train-mlogloss:0.38416	eval-mlogloss:0.33241
[192]	train-mlogloss:0.38410	eval-mlogloss:0.33239
[193]	train-mlogloss:0.38316	eval-mlogloss:0.33111
[194]	train-mlogloss:0.38225	eval-mlogloss:0.33008
[195]	train-mlogloss:0.38183	eval-mlogloss:0.32982
[196]	train-mlogloss:0.38179	eval-mlogloss:0.32978
[197]	train-mlogloss:0.38150	eval-mlogloss:0.32935
[198]	train-mlogloss:0.38147	eval-mlogloss:0.32937

[199]	train-mlogloss:0.38070	eval-mlogloss:0.32851
[200]	train-mlogloss:0.38059	eval-mlogloss:0.32844
[201]	train-mlogloss:0.37894	eval-mlogloss:0.32691
[202]	train-mlogloss:0.37887	eval-mlogloss:0.32686
[203]	train-mlogloss:0.37873	eval-mlogloss:0.32652
[204]	train-mlogloss:0.37812	eval-mlogloss:0.32573
[205]	train-mlogloss:0.37801	eval-mlogloss:0.32545
[206]	train-mlogloss:0.37795	eval-mlogloss:0.32531
[207]	train-mlogloss:0.37788	eval-mlogloss:0.32526
[208]	train-mlogloss:0.37775	eval-mlogloss:0.32518
[209]	train-mlogloss:0.37588	eval-mlogloss:0.32298
[210]	train-mlogloss:0.37574	eval-mlogloss:0.32265
[211]	train-mlogloss:0.37544	eval-mlogloss:0.32223
[212]	train-mlogloss:0.37539	eval-mlogloss:0.32221
[213]	train-mlogloss:0.37534	eval-mlogloss:0.32217
[214]	train-mlogloss:0.37461	eval-mlogloss:0.32146
[215]	train-mlogloss:0.37319	eval-mlogloss:0.31996
[216]	train-mlogloss:0.37242	eval-mlogloss:0.31884
[217]	train-mlogloss:0.37232	eval-mlogloss:0.31861
[218]	train-mlogloss:0.37164	eval-mlogloss:0.31774
[219]	train-mlogloss:0.37064	eval-mlogloss:0.31685
[220]	train-mlogloss:0.37055	eval-mlogloss:0.31680
[221]	train-mlogloss:0.37030	eval-mlogloss:0.31653
[222]	train-mlogloss:0.37025	eval-mlogloss:0.31653
[223]	train-mlogloss:0.37020	eval-mlogloss:0.31652
[224]	train-mlogloss:0.36927	eval-mlogloss:0.31530
[225]	train-mlogloss:0.36850	eval-mlogloss:0.31443
[226]	train-mlogloss:0.36771	eval-mlogloss:0.31340
[227]	train-mlogloss:0.36761	eval-mlogloss:0.31325
[228]	train-mlogloss:0.36694	eval-mlogloss:0.31251
[229]	train-mlogloss:0.36627	eval-mlogloss:0.31183
[230]	train-mlogloss:0.36615	eval-mlogloss:0.31152
[231]	train-mlogloss:0.36606	eval-mlogloss:0.31132
[232]	train-mlogloss:0.36602	eval-mlogloss:0.31135
[233]	train-mlogloss:0.36599	eval-mlogloss:0.31130
[234]	train-mlogloss:0.36575	eval-mlogloss:0.31104
[235]	train-mlogloss:0.36574	eval-mlogloss:0.31105
[236]	train-mlogloss:0.36566	eval-mlogloss:0.31100
[237]	train-mlogloss:0.36551	eval-mlogloss:0.31065
[238]	train-mlogloss:0.36544	eval-mlogloss:0.31065
[239]	train-mlogloss:0.36536	eval-mlogloss:0.31047
[240]	train-mlogloss:0.36454	eval-mlogloss:0.30941
[241]	train-mlogloss:0.36387	eval-mlogloss:0.30853
[242]	train-mlogloss:0.36387	eval-mlogloss:0.30855
[243]	train-mlogloss:0.36382	eval-mlogloss:0.30853
[244]	train-mlogloss:0.36382	eval-mlogloss:0.30854
[245]	train-mlogloss:0.36375	eval-mlogloss:0.30846
[246]	train-mlogloss:0.36374	eval-mlogloss:0.30844

```

[247]   train-mlogloss:0.36302   eval-mlogloss:0.30753
[248]   train-mlogloss:0.36301   eval-mlogloss:0.30753
[249]   train-mlogloss:0.36301   eval-mlogloss:0.30750
[250]   train-mlogloss:0.36296   eval-mlogloss:0.30743
[251]   train-mlogloss:0.36295   eval-mlogloss:0.30740
[252]   train-mlogloss:0.36288   eval-mlogloss:0.30732
[253]   train-mlogloss:0.36282   eval-mlogloss:0.30725
[254]   train-mlogloss:0.36275   eval-mlogloss:0.30721
[255]   train-mlogloss:0.36270   eval-mlogloss:0.30717

[I 2025-12-06 14:55:18,909] Trial 42 pruned. Trial was pruned at iteration 256.

[0]     train-mlogloss:1.07354   eval-mlogloss:1.07319
[1]     train-mlogloss:1.05416   eval-mlogloss:1.05151
[2]     train-mlogloss:1.03165   eval-mlogloss:1.02848
[3]     train-mlogloss:1.01011   eval-mlogloss:1.00678

[I 2025-12-06 14:55:18,974] Trial 43 pruned. Trial was pruned at iteration 4.

[0]     train-mlogloss:0.92985   eval-mlogloss:0.93260

[I 2025-12-06 14:55:19,026] Trial 44 pruned. Trial was pruned at iteration 1.

[0]     train-mlogloss:0.86927   eval-mlogloss:0.86098

[I 2025-12-06 14:55:19,081] Trial 45 pruned. Trial was pruned at iteration 1.

[0]     train-mlogloss:0.77243   eval-mlogloss:0.76395

[I 2025-12-06 14:55:19,134] Trial 46 pruned. Trial was pruned at iteration 1.

[0]     train-mlogloss:0.94533   eval-mlogloss:0.94730

[I 2025-12-06 14:55:19,185] Trial 47 pruned. Trial was pruned at iteration 1.

[0]     train-mlogloss:1.03984   eval-mlogloss:1.03768
[1]     train-mlogloss:1.01179   eval-mlogloss:1.01227
[2]     train-mlogloss:0.96507   eval-mlogloss:0.96325
[3]     train-mlogloss:0.91584   eval-mlogloss:0.91301

[I 2025-12-06 14:55:19,246] Trial 48 pruned. Trial was pruned at iteration 4.

[0]     train-mlogloss:1.07566   eval-mlogloss:1.07553
[1]     train-mlogloss:1.05690   eval-mlogloss:1.05309
[2]     train-mlogloss:1.03869   eval-mlogloss:1.03414
[3]     train-mlogloss:1.01284   eval-mlogloss:1.00680

[I 2025-12-06 14:55:19,306] Trial 49 pruned. Trial was pruned at iteration 4.

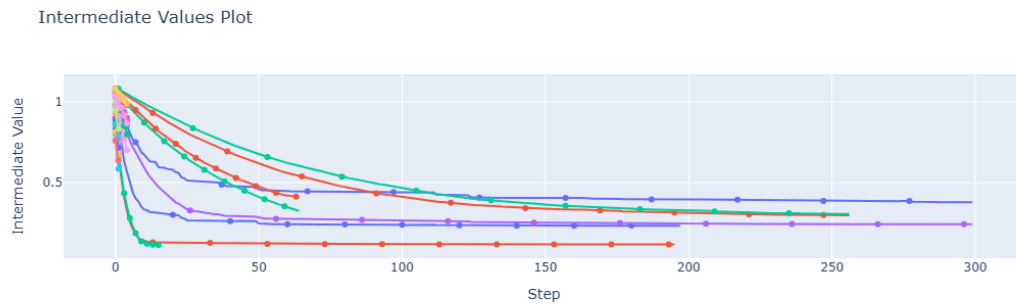
Best trial: {'lambda': 0.03294147801498702, 'alpha': 0.0007672671426894849,
'eta': 0.15847595071800358, 'gamma': 1.0731954455671145e-08, 'max_depth': 4,
'min_child_weight': 8, 'subsample': 0.46617547356234623, 'colsample_bytree':
0.6868533439045852}
Best accuracy: 1.0

```



```
[40]: from optuna.visualization import plot_intermediate_values

# 1. Plot intermediate values during the trials
plot_intermediate_values(study).show()
```



[]:

[]:

[]: